Roadster Thrust System: System, Safety and Security Concept

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An independent systems-engineering, functional-safety (ISO 26262), SOTIF (ISO 21448) and cybersecurity (ISO/SAE 21434) concept study of a cold-gas-thruster "hop" function on a road sports car, inspired by the Tesla Roadster reveal of 15 October 2026.

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Document control

Field	Value
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Revision history

Issue	Date	Change
3.0	18 Oct 2026	Consolidated single document set. Added architecture decision record, assumptions register, requirements database with automated link check, independent review record. Updated E/E architecture for fail-operational hop control (TSR-21 to TSR-23).
2.1	17 Oct 2026	Representative component FMEA and FMEDA, traceability chain, factor-scored TARA and attack tree.
2.0	16 Oct 2026	Industry structure: operating domain, functional and E/E architecture, timing budget, ASIL decomposition, DFA, SOTIF acceptance, cybersecurity and V&V. Corrected fill pressure 600 to 400 bar.
1.0	16 Oct 2026	First architecture and safety analysis.

How this document maps to the standards

Chapter	Work product	Standard reference
1	Executive summary	—
2	Decoded feature and public facts	—
3	Item definition, concept of operations, operating domain	ISO 26262-3 cl. 5; ISO 21448 cl. 5
4	System architecture: logical, functional, E/E, interfaces, modes, timing, design decisions	ISO 26262-4 cl. 6
5	Performance and energy budget	—
6	Hazard analysis and risk assessment, safety goals	ISO 26262-3 cl. 6
7	Functional and technical safety concept	ISO 26262-3 cl. 7; ISO 26262-4 cl. 6
8	Safety analyses: FMEA, FMEDA, FTA, DFA, STPA	ISO 26262-5 cl. 8–9; ISO 26262-9 cl. 7–8
9	SOTIF analysis	ISO 21448 cl. 6–7
10	Cybersecurity concept and TARA	ISO/SAE 21434 cl. 9, 15, Annex G
11	Verification and validation plan	ISO 26262-4 cl. 7–8; ISO 21448 cl. 9–11
12	Traceability	ISO 26262-8 cl. 6
13	Safety case, regulation, review record, open issues	ISO 26262-2 cl. 6
14	Simulation report	Supporting evidence
A–B	Assumptions register, sources	—

Terms and abbreviations

Term	Meaning
ASIL	Automotive safety integrity level, A (lowest) to D (highest); B(D) means ASIL B after decomposition of a D goal
CAL	Cybersecurity assurance level (ISO/SAE 21434), 1 to 4
COPV	Carbon-fibre overwrapped pressure vessel
DC	Diagnostic coverage
DFA	Dependent failure analysis
FHA	Functional hazard assessment (aviation, ARP4761A)
FIT	Failures in time: one failure per 10^9 hours
FSR, TSR	Functional and technical safety requirement
FTTI	Fault tolerant time interval: time from fault to hazard if nothing reacts
HIL	Hardware-in-the-loop testing
SPFM, LFM, PMHF	Single-point fault metric, latent fault metric, probabilistic metric for hardware failures
SOTIF	Safety of the intended functionality: hazards with nothing broken
STPA	System-theoretic process analysis
TARA	Threat analysis and risk assessment
TCU	Thrust control unit (the command-channel computer)

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1. Executive summary

The Roadster's "flying" feature is a cold-gas thruster system developed with SpaceX, paired with a fan-driven suction system. It gives powered hops of a few seconds, not sustained flight. This study shows the concept can be made acceptably safe only with four architectural measures: two-channel arming with two master valves in series, a "recover, then cut" response to faults in flight, fail-operational control during a hop, and pre-flight checks of every valve and of the gas reserve.

Key findings

#	Finding	Evidence
1	It is a hop, not a flight: about 244 kg of air at about 400 bar holds a 1,850 kg car in a hover for 6.7 s, and one reference hop uses 55% of it	Sim 1
2	The gas storage is the largest stored-energy hazard: about 60 MJ of burst energy (about 14 kg TNT), and a full discharge can chill the gas towards −130 °C	Sim 1
3	Cutting all gas mid-air is always fatal at hop heights. Isolating only the failed line, then recovering, cut bad outcomes after a stuck-open valve from 100% to 0–3%	Sims 2, 4
4	The fault tolerant time interval for a stuck-open valve is about 200 ms; detection and isolation are budgeted at 20 ms	Sim 3
5	A stuck-closed valve during the roll (1.10–1.95 s) almost always ends badly; the same fault outside the roll is always safe	Sim 4, viewer
6	Valves that close on power loss are safe on the ground but drop the car in the air. Three requirements (TSR-21 to TSR-23) make hop control fail-operational	Component FMEA
7	SG1 (no unintended thrust) meets the ASIL D hardware targets with representative rates: SPFM 100%, LFM 98.7%, PMHF 0.006 FIT. Common-cause faults dominate what risk remains	Sim 7
8	A faked track zone is the one attack path that defeats both channels at once, because both trust the same zone data; an independent zone check closes it	TARA, attack tree
9	Per-hop risk of a bad landing from faults is 3.3×10^{-7} against a proposed 10^{-7} target, and the SOTIF residual is bounded only at 1.4×10^{-3} per hop	Sims 4, 6
10	There is no regulatory path yet: Tesla lists the Rocket trim as not approved for public road use	Chapter 13

Status at a glance

Item	Count	State
Hazardous events	11	Rated; 3 at ASIL D
Safety goals	8	3 ASIL D, 2 ASIL C, 3 ASIL B
Functional safety requirements	14	13 in scope; 1 allocated to the steer-by-wire item
Technical safety requirements	23	All allocated and with a verification method
Verification items	25	11 with analysis or simulation evidence, 2 showing a gap, 12 not started
Failure modes analysed	31 component-level	23 at action priority H
Threat scenarios	9	All at risk value 3 or below after 13 controls
Traceability check	Safety and security chains, both directions	Passes (`tools/check_traceability.py`)

What would block release

1. SOTIF residual risk is not yet demonstrated (G3 in the safety case).
2. Stuck-closed valve authority (TSR-17) and the recovery planner (TSR-20) show gaps in simulation.
3. No physical test evidence exists; every number rests on the assumptions in Appendix A.

2. Decoding the feature

Tesla has no single product name for the flying feature. It appears under four names, plus a separate suction system that does the upside-down driving.

Name	What it refers to	Status
SpaceX package	Musk's 2018 name for an option with about 10 small thrusters arranged around the car	Public since June 2018 [B8]
A71	Reported internal code name for the Tesla–SpaceX cold-gas thruster system	Reported; not confirmed by Tesla [B6]
Thrust trim	"Race ready", 2 seats, rear-mounted cold-gas thrusters for 0–60 mph under 1 s	$250,000 reservation [B1, B2]
Rocket trim	"Air ready", 1 seat, 10 thrusters around the car for forward, upward and rotational moves; Tesla says it "can fly"	Invite only; not approved for public road use [B2]
Multi-mode suction system	Fans and a ground-sealing skirt that pull air from under the car, strong enough to hold it upside down when stationary	All three trims [B2]

What Tesla has disclosed

- 10 thrusters fed by 600 litres of rechargeable compressed air (L. Moravy) [B1]; 250 kg of compressed gas reported by one outlet [B3].
- Every usable void in the car is packed with carbon-fibre overwrapped pressure vessels (COPVs) [B1].
- All-electric actuators make micro-adjustments to stabilise the car in flight [B2].
- Suction allows cornering up to 2G [B3]. 0–60 mph under 1.6 s (Performance), under 0.9 s (Thrust), under 0.59 s (Rocket) [B4, B5].
- Steer-by-wire yoke, Track Mode, magnesium, titanium and carbon-fibre construction [B2].
- The demonstration car was remotely operated with nobody inside. It drove up an inverted ramp, spun and landed, then cleared a Cybercab, Cybertruck, Semi, Model Y and Model 3 [B1].
- Production is planned for 2028 [B1, B2].

Everything else in this document is an assumption, numbered A-01 to A-16 in Appendix A.

3. Item definition and concept of operations

3.1 Item and boundary

The item is the thrust and suction function: gas supply and storage, distribution, thruster array, suction, the sensing and computers that control them, the arming interface and the protection measures specific to it (subsystems S2–S6 and S8–S11, with their links to S1 and S7). The vehicle platform, steering and airbags are outside the item and appear as interfaces. Thrust functions are allowed only inside a bounded operating domain; everywhere else the car is in Road-safe mode, where thrust is physically impossible.

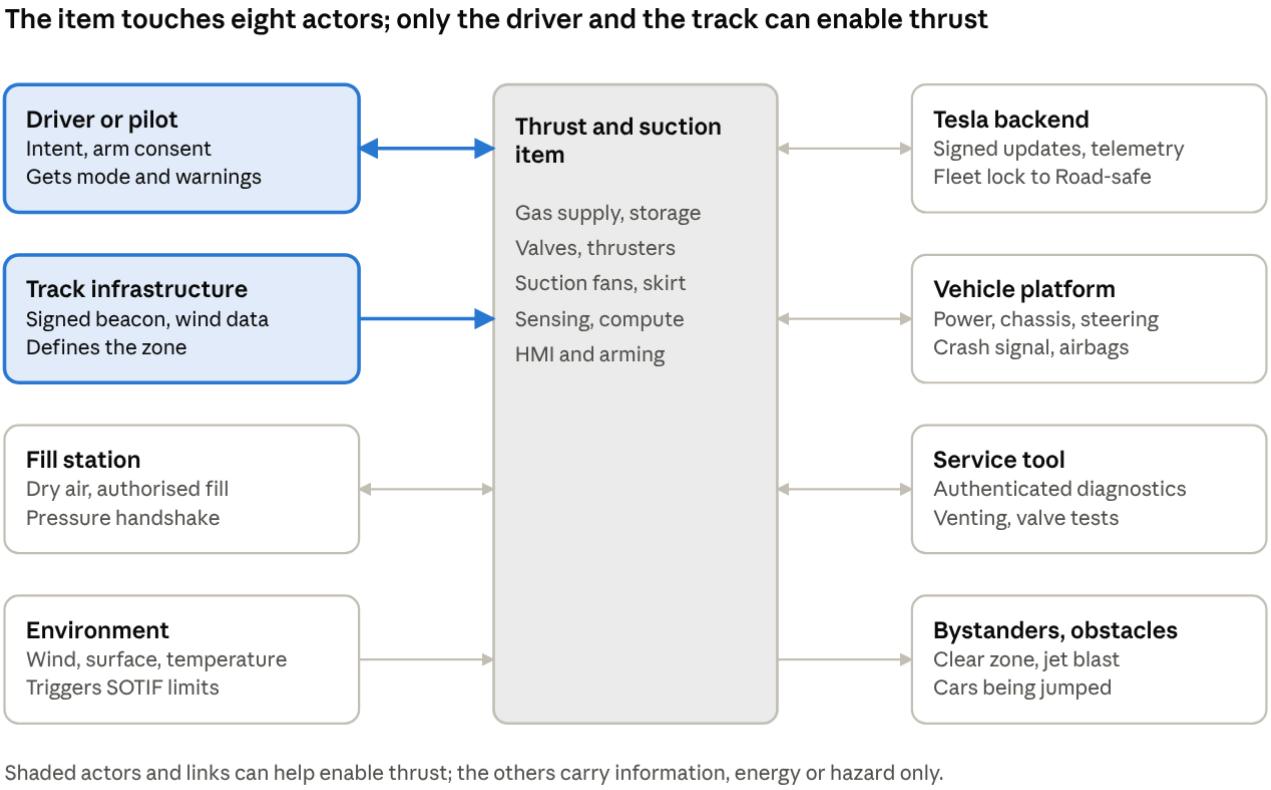


Figure 1. System context: the item, its eight actors and the only two that can contribute to enabling thrust.

3.2 Use cases

ID	Use case	Who	Mode	Trim
UC1	Normal road driving	Owner	Road-safe	All
UC2	Track driving with suction	Owner	Track	All
UC3	Straight-line boost launch	Owner	Boost	Thrust, Rocket
UC4	Hop or roll manoeuvre	Invited pilot	Hop	Rocket
UC5	Refill the air tanks	Fill-station operator	Road-safe	Thrust, Rocket
UC6	Service, inspect and vent the tanks	Technician	Vent and maintain	Thrust, Rocket
UC7	Crash in any mode	—	Crash	All
UC8	Remote-operated test (development only)	Test engineer	Hop	Prototype

3.3 Operating domain for thrust functions

Dimension	Allowed	Monitored by
Location	Authorised track or closed site inside a geofence	Satellite positioning plus signed track beacon
People	Nobody inside the clear zone (proposed 50 m for Hop, 20 m for Boost)	Cameras and site procedure
Surface	Dry, sealed, paved; no loose gravel	Suction plenum pressure, cameras
Wind	Sustained under 8 m/s, gusts under 12 m/s	Site anemometer feed
Temperature	−10 to +40 °C ambient	Vehicle sensors
Weather	No rain, snow or ice; daylight	Rain sensor, site procedure
Occupants	Belted; harness and helmet in Rocket trim	Seat sensors
Vehicle state	Gas above reserve; all pre-arm checks passed; no active faults	Mode manager
Manoeuvre	Hop apex up to 6 m; Boost in a straight line only	Trajectory planner

3.4 Foreseeable misuse

- Firing thrusters on public roads or in car parks.
- Spoofing or disabling the geofence or beacon (see TS3, TS4 in chapter 10).
- Refilling with wet air, oxygen-enriched air or another gas at an unapproved station.
- Carrying a passenger in the single-seat Rocket trim, or loose cargo during a hop.
- Letting spectators stand inside the clear zone.

4. System architecture

The architecture rests on two rules. On the ground, thrust is impossible unless two independent channels agree, so the default state is gas isolated and no thrust. In the air, no single fault may remove thrust, so the car can always level itself and land. The rest of this chapter shows how the subsystems, functions, electronics, interfaces and modes deliver both rules, and why each major design decision was taken. This is a reference architecture inferred from public disclosures; Tesla has not published its own.

4.1 Architecture principles

ID	Principle	Driven by
P1	Fail closed on the ground: every gas valve is normally closed and needs active power to open	SG1, SG2
P2	Fail operational in the air: during a hop no single fault closes the gas or stops control for at least 5 s	SG6
P3	Two independent, dissimilar channels must both agree before gas can reach a nozzle	SG1
P4	The thrust domain is isolated: nothing outside it can command thrust	SG1, CG1
P5	Attitude before altitude: on any in-flight fault, level the car first, then descend	SG6
P6	Never start what cannot be finished after one fault (gas reserve, lift margin)	SG6

4.2 Logical architecture

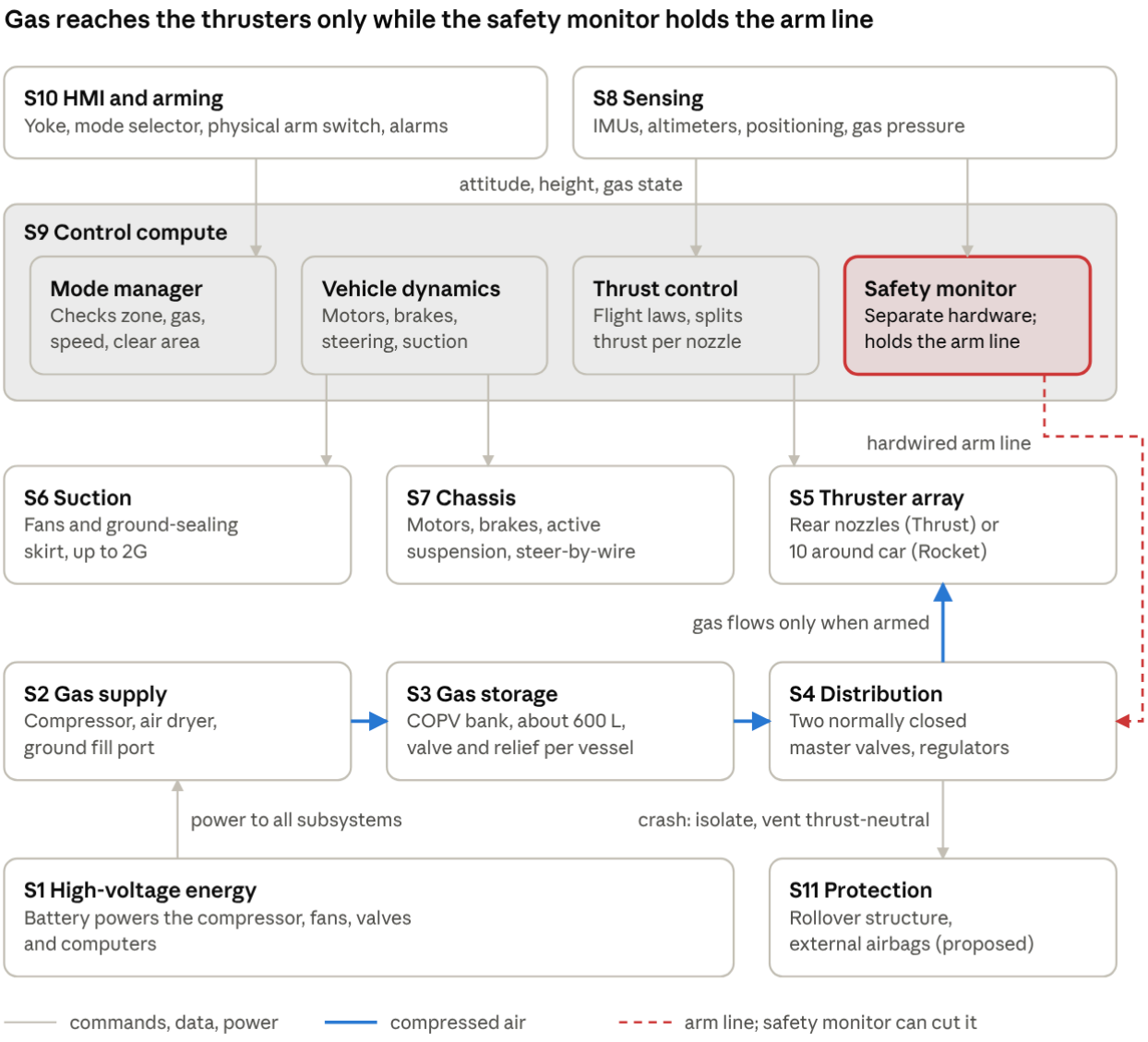


Figure 2. Logical architecture: eleven subsystems in six layers (energy, gas, effectors, sensing, control, protection), with the gas path and the hardwired arm line.

ID	Subsystem	Content	Technical safety requirements allocated
S1	High-voltage energy	Battery pack and DC-DC converters powering compressor, fans, valves and compute	TSR-21
S2	Gas supply	Onboard compressor, air dryer, ground fill port, check valves	TSR-19
S3	Gas storage	COPV bank (about 600 L at 400 bar), per-vessel isolation and relief, pressure and temperature sensing	TSR-05, TSR-06, TSR-07, TSR-10, TSR-18
S4	Distribution	Two normally closed master valves in series, regulators, manifold, line valves	TSR-01, TSR-03, TSR-07, TSR-09, TSR-16, TSR-18, TSR-21
S5	Thruster array	Rear nozzles (Thrust) or 10 perimeter nozzles (Rocket) with fast electric valves and heaters	TSR-09, TSR-15, TSR-16, TSR-17, TSR-18
S6	Suction	Multi-mode fans, ground-sealing skirt, plenum pressure sensors	TSR-08

ID	Subsystem	Content	Technical safety requirements allocated
S7	Chassis	Motors, brakes, steer-by-wire, active suspension, active aero	TSR-08
S8	Sensing	Three IMUs, altimeters, satellite positioning, wheel speeds, cameras	TSR-02, TSR-03, TSR-05, TSR-13
S9	Control compute	Mode manager, vehicle dynamics controller, thrust control unit, independent safety monitor	TSR-01, TSR-02, TSR-03, TSR-04, TSR-09, TSR-10, TSR-11, TSR-12, TSR-16, TSR-17, TSR-20, TSR-21, TSR-22, TSR-23
S10	HMI and arming	Yoke, mode selector, physical arm switch, displays and alarms	TSR-02
S11	Protection	Restraints, rollover structure, external airbags, thrust-neutral emergency venting	TSR-07, TSR-14

4.3 Functional architecture

Fourteen functions make up the item. Each is allocated to a subsystem and inherits the highest ASIL of the safety goals it supports.

ID	Function	Allocated to	Supports	ASIL
F1	Capture driver intent and arming consent	S10	SG1	D
F2	Confirm zone authorisation from two independent sources	S8, S9	SG1, SG7	D
F3	Measure gas state and predict usable reserve	S3, S9	SG2, SG6	D
F4	Plan the manoeuvre and check it can be finished after one fault	S9	SG6	B
F5	Estimate attitude, height and position	S8	SG6	B
F6	Compute flight control laws	S9	SG5, SG6	C
F7	Allocate thrust to nozzles, torque first	S9	SG5, SG6	C
F8	Actuate thruster valves	S5	SG1, SG5	D
F9	Isolate gas (master and line valves)	S4, S9	SG1, SG2, SG6	D
F10	Monitor, arbitrate and command recovery; run the backup hop law	S9	SG1, SG6	D
F11	Generate downforce	S6	SG4	C
F12	Protect occupants (restraints, external airbags)	S11	SG8	B
F13	Store and refill gas	S2, S3	SG2	D
F14	Vent gas without net thrust	S3, S4	SG2	D

4.4 E/E and software architecture

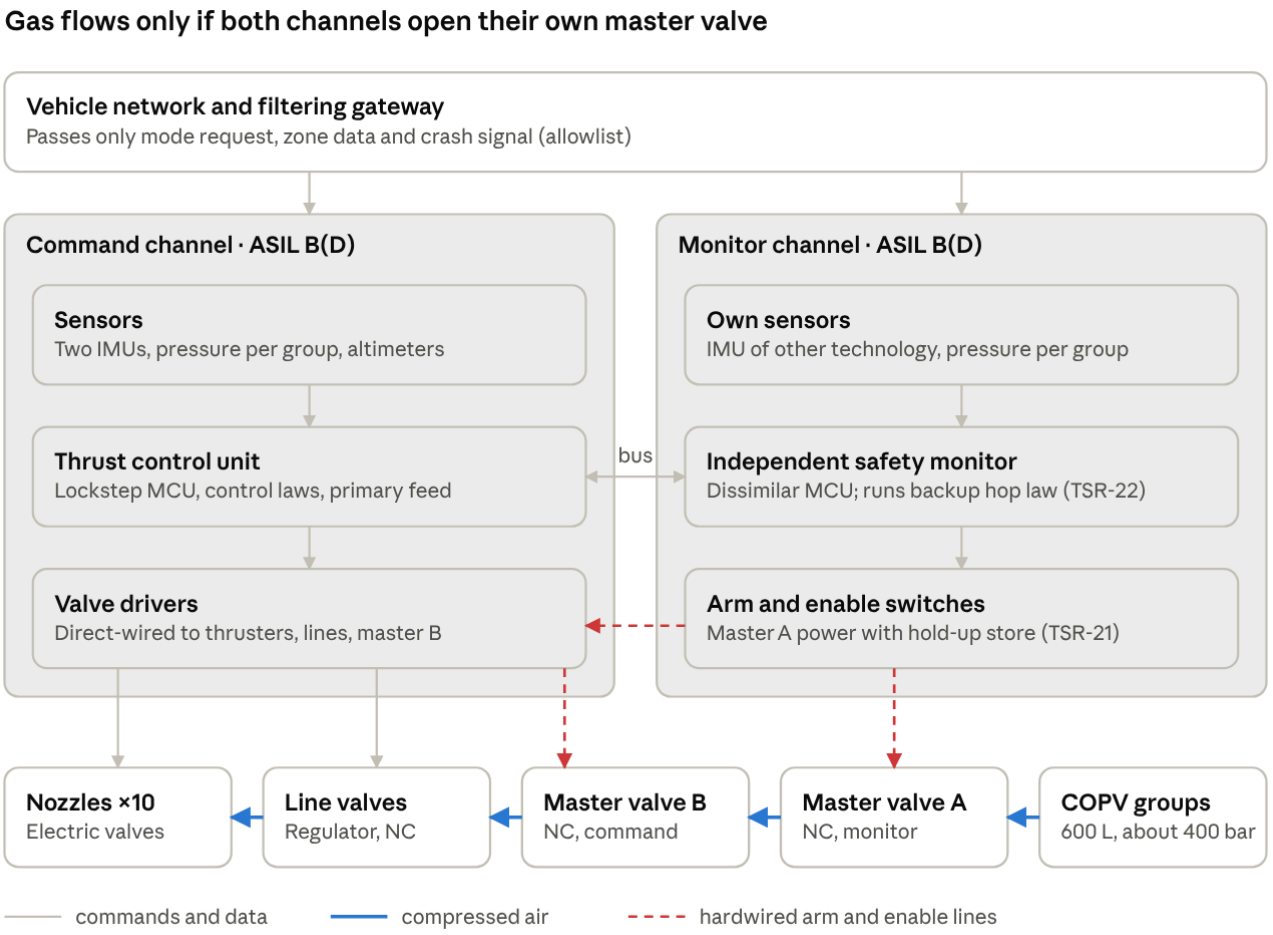


Figure 3. E/E architecture: two channels each own one of the two master valves in series; the monitor also owns the valve-driver supply, the hold-up store and the backup hop law.

Element	Command channel	Monitor channel
Computer	Thrust control unit: lockstep multicore microcontroller	Independent safety monitor: dissimilar microcontroller from a different supplier
Software	Partitioned real-time OS: control laws, allocation, recovery planner, diagnostics; separate logging partition	Small bare-metal program: envelope checks, arm logic, recovery command, backup attitude-hold and descent law
Sensors	Two IMUs, one pressure sensor per gas group, altimeters	Own IMU of a different technology, own pressure sensor per group
Gas valves owned	Master valve B, line and thruster valves (direct-wired)	Master valve A (with hold-up store), enable of all valve-driver supplies
Power	Primary low-voltage feed	Secondary feed with local buffer
ASIL after decomposition	B(D) for SG1; C for SG5; B for SG6	B(D) for SG1; B for SG6 recovery and backup control

ASIL decomposition of SG1. SG1 (ASIL D) is decomposed into the command channel at ASIL B(D) and the monitor channel at ASIL B(D). The decomposition is valid only if the dependent failure analysis (section 8.5) shows the channels are sufficiently independent; the FMEDA in section 8.3 shows common cause is the dominant residual, which is why that analysis is decisive.

Network. Thrust messages run on a private, authenticated bus between the two channels. The vehicle network can send only a mode request, zone data and the crash signal, through a filtering gateway. No cellular, Wi-Fi or diagnostic path reaches the thrust domain in production.

Fail-safe and fail-operational hardware. Every gas valve is normally closed and spring-return, so loss of power on the ground always closes it (P1). In Hop mode, dual-wound coils, a second driver per master valve and a local hold-up energy store keep the master valves open through any single loss for at least 5 s, and closing them in flight takes an explicit command (P2, TSR-21).

4.5 Interfaces

ID	Signal	From → to	Type	Rate	Max latency	ASIL	Protection
IF-01	Arm enable A	Safety monitor → master valve A switch	Hardwired, energise-to-open	Continuous	5 ms to close	B(D)	Dedicated wire, read-back, short detection
IF-02	Arm enable B	Thrust control unit → master valve B switch	Hardwired, energise-to-open	Continuous	5 ms to close	B(D)	As IF-01
IF-03	Thrust command	Thrust control unit → valve drivers	Direct-wired, or two independent buses (TSR-23)	1 kHz	2 ms	C	Counter, CRC, timeout, authentication
IF-04	Driver enable	Safety monitor → valve-driver supply	Hardwired	Continuous	5 ms	B(D)	Read-back
IF-05	Zone authorisation	Positioning and beacon → mode manager	Vehicle bus via gateway	10 Hz	200 ms	D (two sources)	Two sources must agree; signed beacon; independent check in the monitor (CC-13)
IF-06	Attitude and rates	IMUs → both computers	Private bus or direct SPI	1 kHz	2 ms	B	Three-way voting, plausibility
IF-07	Gas pressure and temperature	Sensors → both computers	Analogue or private bus	500 Hz	5 ms	D	Two dissimilar sensors per group
IF-08	Suction plenum pressure	Sensors → vehicle dynamics controller	Vehicle bus	200 Hz	10 ms	C	Range and rate checks
IF-09	Mode request	HMI → mode manager	Vehicle bus via gateway	Event	50 ms	D	Physical arm switch plus two-step confirm
IF-10	Crash signal	Airbag controller → both computers	Hardwired plus bus	Event	10 ms	D	Redundant paths
IF-11	Line isolation command	Thrust control unit → line valves	Direct-wired	Event	20 ms	B	Read-back of valve position

4.6 Operating modes

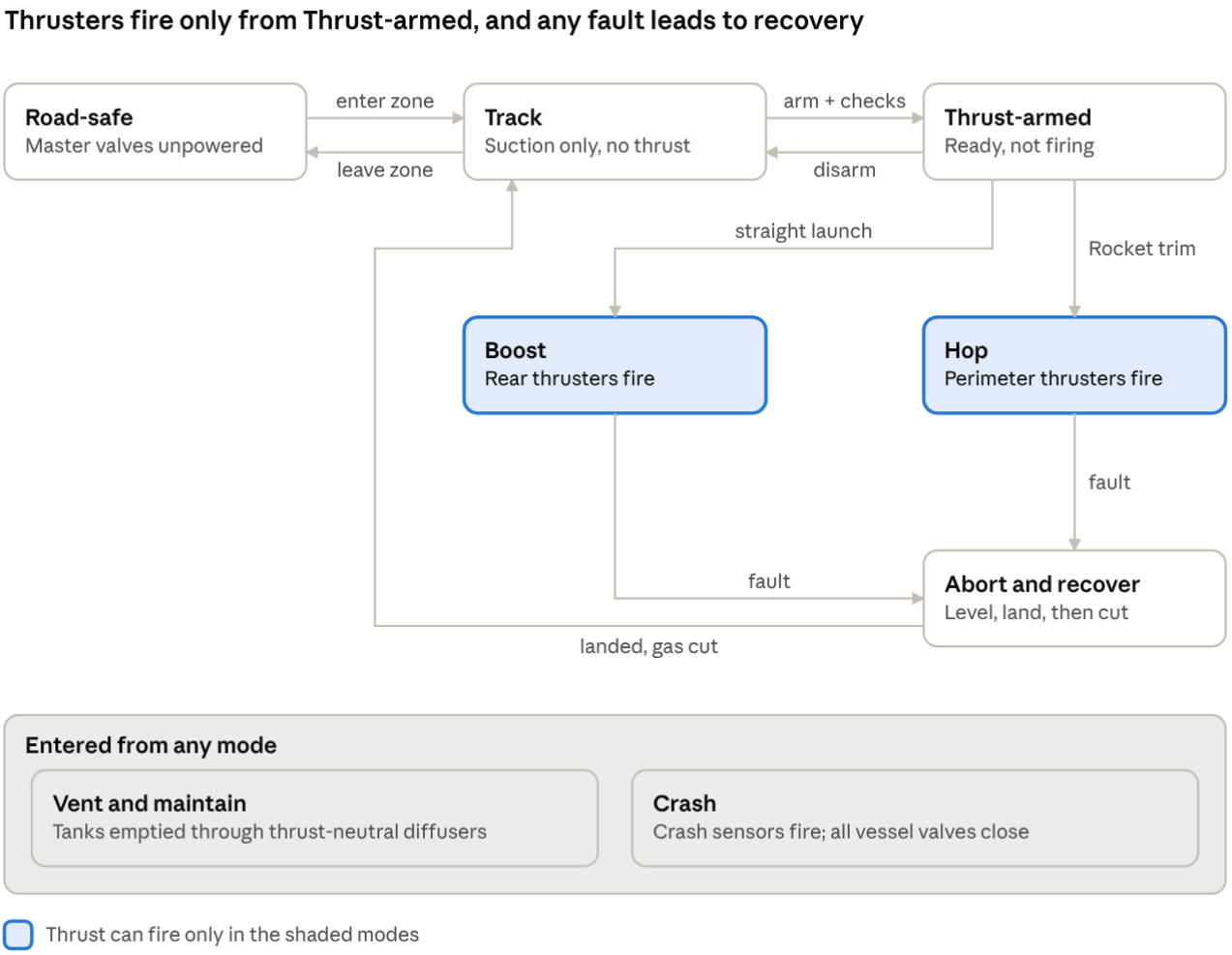


Figure 4. Operating modes: thrusters can fire only in Boost and Hop, entered only from Thrust-armed; any fault leads to recovery.

Mode	Thrust	Entry conditions
Road-safe	Physically impossible: master valves unpowered	Default at power-on and on public roads
Track	No; suction allowed	Driver request inside an authorised track zone
Thrust-armed	Ready, not firing	Track mode, physical arm switch, gas reserve check passed, clear zone confirmed
Boost	Rear thrusters	Armed, wheels on ground, straight-line launch
Hop	Perimeter thrusters	Armed, Rocket trim, manoeuvre passes the reserve and lift-margin checks
Abort and recover	Remaining gas used to level the car, then land	Any fault during Boost or Hop
Vent and maintain	No; tanks emptied through diffusers	Service, or a fault that needs gas removed
Crash	No; all vessel valves closed	Crash sensors fire, in any mode

4.7 Timing budget

The tightest timing is the stuck-open valve in flight: the FTTI is about 200 ms with 9 kN lift groups (Sim 3), against a 20 ms detection-to-isolation budget. For SG1 the safe state is held continuously, so no reaction time applies.

Goal	Fault	FTTI	Detection	Decision	Actuation	Budget	Basis
SG6	Stuck-open thruster valve in flight	200 ms	10 ms (line pressure vs command, IMU residual)	2 ms	8 ms line valve closes	20 ms (ceiling 50 ms)	Sim 3
SG6	Leak or pressure loss in flight	Depends on leak rate	20 ms (pressure decay vs model)	2 ms	Next control cycle	25 ms	Sim 4
SG6	Thrust control unit lost in flight	Under 200 ms (estimate)	10 ms (watchdog)	5 ms	Backup law takes over	50 ms	TSR-22
SG5	Asymmetric thrust in launch	About 100 ms (estimate)	5 ms (yaw rate vs model)	1 ms	8 ms master valve closes	15 ms	Estimate
SG4	Suction loss at high cornering load	About 150 ms (estimate)	10 ms (plenum pressure)	5 ms	Speed and cornering limits	100 ms	Estimate
SG2	Over-pressure while refilling	Seconds	100 ms	10 ms	Stop fill; mechanical relief as backup	1 s	Estimate

Loop rates: attitude control and allocation 1 kHz; monitor envelope checks 1 kHz; mode manager 100 Hz; zone authorisation 10 Hz.

4.8 Architecture decision record

ID	Decision	Alternatives considered	Rationale	Consequence
AD-01	Two normally closed master valves in series, each powered by a different channel	One master valve with redundant drivers	Enables SG1 decomposition into B(D) + B(D); no single valve or driver fault releases gas	Extra valve mass and pressure drop; leak-down test needed to keep faults from going latent
AD-02	Dissimilar safety monitor from a different supplier	Second identical lockstep computer	Reduces common-cause faults, the dominant residual in the FMEDA	Two code bases and tool chains to maintain
AD-03	Recover, then cut, in flight	Cut all gas on any fault	Sim 2: cutting mid-roll lands the car on its roof; recovery lands it upright	Monitor logic must know the flight phase; cutting stays the safe state on the ground
AD-04	Fail-operational hop control: hold-up store, dual-wound coils, backup law	Rely on fail-closed valves	Component FMEA: loss of power or a computer mid-hop otherwise drops the car	More hardware in the monitor channel; must not weaken SG1 on the ground
AD-05	Isolated thrust domain behind an allowlist gateway	Thrust functions on the shared vehicle network	Removes remote attack paths to a severe damage scenario	Updates to the thrust domain only with tanks empty and car present
AD-06	Gas split into at least two vessel groups with own isolation valves	One manifolded bank	A leak or stuck valve costs part of the reserve, not all of it	More valves and sensors
AD-07	Thrust-neutral emergency venting through paired diffusers	Single vent	Venting must not push the car, especially after a crash	Packaging of opposed diffusers
AD-08	Reserve check before every manoeuvre, offering a lower manoeuvre instead of refusing	Fixed minimum fill	Sim 4: without it, 21% of fault-free random hops starve	Availability cost at low fill
AD-09	Monitor confirms the zone independently (inertial dead-reckoning from the last trusted beacon fix)	Both channels use the same zone data	Attack tree: a faked zone otherwise defeats both channels at once	Beacon infrastructure must sign fixes

5. Performance and energy budget

The budget turns the disclosed figures into the physical limits every later chapter depends on. All values come from Sim 1 (chapter 14) using the assumptions in Appendix A; the most sensitive is the fill pressure (A-05), which Tesla has not disclosed.

Quantity	Value	How it is obtained
Gas volume	600 L	Disclosed [B1]
Fill pressure	About 400 bar	Assumed (A-05); 600 bar would need a stronger vessel class than reported
Gas mass at fill	About 244 kg	Real-gas density at 400 bar, 293 K (compressibility Z ≈ 1.17); consistent with the 250 kg figure in [B3]
Weight to hold in a hover	About 18.1 kN	1,850 kg vehicle (A-01)
Specific impulse of the jets	69 s at fill, falling to 48 s as the gas cools	Ideal nozzle expansion of air from the regulated supply
Hover endurance, full tanks	6.7 s	Adiabatic blowdown until supply pressure falls below the regulator minimum
Gas left unusable at the end	About 40 kg	Tank pressure falls to about 28 bar
Gas temperature at the end of a full discharge	About 142 K (−131 °C)	Adiabatic expansion in the tanks
Reference hop (4.5 m apex, 360° roll, soft landing)	134 kg, 4.15 s, 55% of the fill	Sim 1, Sim 2
Minimum fill to start the reference hop with a 1.2 reserve factor	86%	Sim 4
Launch, 0–60 mph in 0.9 s (Thrust trim)	32–46 kg of gas	Thrust needed beyond tyre grip, 13–19% of the fill
Launch, 0–60 mph in 0.59 s (Rocket trim)	46–67 kg of gas	19–27% of the fill
Stored energy if a full bank bursts	About 60 MJ (about 14 kg of TNT)	Isentropic expansion energy of the stored gas
Energy to refill, ideal	About 34 kWh	Isothermal compression; a real compressor needs two to three times more

What the budget means for the architecture.

1. *It is a hop, not a flight.* A full bank gives under seven seconds of hover, and one reference hop uses more than half of it. There is no gas for a second attempt, so the reserve check (TSR-10) and recovery on the remaining gas (TSR-11) are essential, not optional.
2. *The vessels are a stored-energy hazard that can be released in milliseconds*, unlike the battery's chemical energy, which is released over seconds to minutes. SG2 is ASIL D for this reason, and vessel placement, shielding and thrust-neutral venting (AD-07) are architecture decisions, not packaging details.
3. *Cold is a design load.* Valves, seals and liners must work at the coldest gas temperature of a full discharge (TSR-18), and refill air must be dry (TSR-19), or ice can jam a valve when it is needed most.
4. *The car cannot refill itself quickly.* An on-board compressor powered from the traction battery would take far longer than a station refill; the operating concept assumes a fill station at the track (UC5).

6. Hazard analysis and risk assessment

6.1 Method

Hazardous events are rated with ISO 26262-3 severity (S), exposure (E) and controllability (C) classes. Because a hop is closer to a short flight than to driving, each in-flight failure condition is also classified the aviation way (ARP4761A functional hazard assessment) and given a probability target per hop. Where the two methods disagree, the stricter result governs.

Rating rules used throughout:

- **Severity.** Any event that can kill an occupant or a bystander is S3. A hop from 4.5 m that ends inverted or at more than 5 m/s vertical speed is treated as potentially fatal.
- **Exposure.** Public-road situations are E4. Track use with the thrusters armed is E3 for Boost, which any Thrust-trim owner can use, and E2 for Hop, which needs the invite-only Rocket trim at an authorised site. The inverted-ramp stunt is E1.
- **Controllability.** No driver can correct an unexpected thrust event or an in-flight fault within the time available (C3), except low-speed manoeuvring, where braking can partly help (C2).

6.2 Hazardous events

ID	Hazardous event	Situation	S	E	C	ASIL	Goal
HE1	Unintended thruster firing	Driving on a public road	S3	E4	C3	D	SG1
HE2	Unintended thruster firing	Parking or low speed near people	S3	E4	C2	C	SG1
HE3	COPV rupture or uncontrolled venting	Any driving or parking	S3	E4	C3	D	SG2
HE4	Loss of steering	Any driving (steer-by-wire)	S3	E4	C3	D	SG3
HE5	Sudden loss of suction at high cornering load	Track mode, near 2G	S3	E3	C3	C	SG4
HE6	Asymmetric thrust during launch	Boost, 3-4.6 g straight-line launch	S3	E3	C3	C	SG5
HE7	Loss of thrust mid-manoeuvre (gas used up, leak, icing, valve closure)	Hop	S3	E2	C3	B	SG6
HE8	Stuck-open, stuck-closed or wrong-direction thruster	Hop	S3	E2	C3	B	SG6
HE9	Jet blast or car strikes a bystander	Hop or Boost with a person in the area	S3	E2	C3	B	SG7
HE10	Unintended external airbag deployment	Highway driving	S2	E4	C2	B	SG8
HE11	Loss of suction while inverted	Inverted-ramp stunt	S3	E1	C3	A	SG6

6.3 Functional hazard assessment for flight phases

Failure condition	Phase	Effect	Class	Target per hop
Total loss of thrust	Climb, apex, roll	Fall from up to 6 m, possibly inverted	Catastrophic	Below 10^{-7}
Uncontained asymmetric thrust	Any airborne phase	Uncontrolled roll; inverted landing	Catastrophic	Below 10^{-7}
Loss of attitude information	Any airborne phase	Uncontrolled attitude	Catastrophic	Below 10^{-7}
Single stuck-closed nozzle	Roll	Reduced roll braking; possible overshoot	Hazardous	Below 10^{-5}
Partial loss of lift	Descent	Hard landing at 3–5 m/s	Major	Below 10^{-3}
Lateral drift beyond 3 m	Any	Car leaves the landing area	Major	Below 10^{-3}

6.4 Safety goals

ID	Safety goal	ASIL	Safe state	FTTI	From
SG1	Prevent thrust unless armed in an authorised zone with a valid driver command	D	Master valves closed (Road-safe)	Preventive	HE1, HE2
SG2	Prevent COPV rupture or uncontrolled venting from harming anyone	D	Vessels isolated; thrust-neutral venting	Seconds (fill over-pressure)	HE3
SG3	Maintain steering control	D	Redundant steer-by-wire, degraded mode	Per steering item	HE4
SG4	Prevent a loss of downforce greater than the chassis can handle	C	Speed and cornering limits, warning	About 150 ms (estimate)	HE5
SG5	Prevent asymmetric thrust during a launch	C	Cut thrust (safe on the ground)	About 100 ms (estimate)	HE6
SG6	Ensure any started manoeuvre can be finished and landed upright after one fault	B + FHA target 1e-7 per hop	Level the car, land, then cut	200 ms (Sim 3)	HE7, HE8, HE11
SG7	Prevent jet blast or the car striking a person	B	Refuse to arm	Preventive	HE9
SG8	Prevent external airbag deployment outside Hop mode	B	Airbags disarmed	Preventive	HE10

6.5 Rationale for difficult ratings

Why SG6 is ASIL B with a quantitative target. By the ISO 26262 tables, a fatal loss of control during a hop is S3, C3 and E2, which yields ASIL B. That rating reflects how rarely the hop situation occurs across the whole fleet, not how dangerous each hop is. For the person in the car, a hop is a flight. An ASIL B rating on its own would allow a per-hop risk far above what aviation accepts for a catastrophic failure. This study therefore keeps ASIL B for the development process and adds the FHA target of 10^{-7} per hop as a quantitative requirement, checked in Sim 6 and section 9.3. If the operator markets the hop as a routine feature (E3 or E4), SG6 rises to ASIL C or D.

Why HE11 is merged into SG6. Loss of suction while inverted (ASIL A on its own) has the same safe state as an in-flight fault: level the car and land. It is handled by SG6's recovery requirements.

Why SG3 is outside the item. Steering is a separate steer-by-wire item with its own ASIL D goal. It appears here only as an interface; FSR-3.1 records the allocation.

Why HE1 and HE2 share SG1. They are the same malfunction in different situations; the goal takes the worse rating (ASIL D).

7. Functional and technical safety concept

7.1 Functional safety requirements

Each safety goal is refined into functional safety requirements that say *what* the system must do, without naming hardware.

ID	Functional safety requirement	Goal	ASIL
FSR-1.1	Keep the gas path to the nozzles closed unless two independent channels both authorise it	SG1	D
FSR-1.2	Authorise thrust only with driver consent, an authorised zone confirmed by two independent sources and passed pre-arm checks	SG1	D
FSR-1.3	Accept thrust commands only from authenticated sources inside the thrust domain	SG1	D
FSR-2.1	Contain the stored gas under all foreseeable pressure, temperature and impact loads	SG2	D
FSR-2.2	On crash or fire, isolate every vessel and vent only without net thrust	SG2	D
FSR-3.1	Maintain steering control (allocated to the steer-by-wire item; outside this item)	SG3	D
FSR-4.1	Detect loss of downforce and limit speed and cornering before grip is exceeded	SG4	C
FSR-5.1	Detect asymmetric thrust during a launch and stop it	SG5	C
FSR-6.1	Start a manoeuvre only when the remaining gas can finish it after one fault	SG6	B
FSR-6.2	Respond to any in-flight fault with recovery to an upright landing, never with loss of all thrust	SG6	B
FSR-6.3	Isolate a faulty thruster path and compensate with the others	SG6	B
FSR-6.4	Estimate attitude, height and gas state with tolerance to one sensor fault	SG6	B
FSR-7.1	Refuse to arm unless the clear zone is empty and wind is within limits	SG7	B
FSR-8.1	Arm external airbags only in Hop mode above 1.5 m	SG8	B

7.2 Safety mechanisms

Mechanism	Detects	Reaction	Goal	Covered by
Two-channel arming with master valves in series	Wrong arming decision by either computer	The other channel's valve stays closed	SG1	TSR-01, TSR-02
Power-on leak-down test of master valves	Latent stuck-open or leaking master valve	Thrust locked out	SG1, SG2	TSR-03
Pre-arm valve function test	Stuck-open or stuck-closed thruster, line or master valve	Arming refused	SG1, SG6	TSR-16
Line pressure against command, IMU residual	Stuck-open thruster valve in flight	Line isolated within 20 ms, others compensate	SG5, SG6	TSR-09, TSR-15
Gas reserve prediction	Too little gas to finish after one fault	Lower manoeuvre offered or recovery started	SG6	TSR-10, TSR-11
Monitor envelope check	Attitude, rate or height outside the planned envelope	Recovery commanded, not gas cut	SG6	TSR-12, TSR-20
IMU voting with dissimilar check	Single attitude-sensor fault	Faulty sensor masked	SG6	TSR-13
Fail-operational hop hardware	Power, coil, switch or computer loss mid-hop	Master valves held open; backup law takes over	SG6	TSR-21, TSR-22, TSR-23
Plenum pressure monitoring	Suction loss	Speed and cornering limits, warning	SG4	TSR-08

Mechanism	Detects	Reaction	Goal	Covered by
Crash isolation	Crash or fire	All vessel valves close; thrust-neutral venting	SG2	TSR-07
Dual dissimilar pressure sensing	Pressure sensor fault	Faulty reading masked; fill or hop refused	SG2, SG6	TSR-05
Fill-coupling dew-point check	Wet refill air	Fill refused	SG2	TSR-19
Airbag arming window	Deployment outside Hop mode	Airbags stay disarmed	SG8	TSR-14

7.3 Degradation and warning concept

Situation	System response	Driver information
Any fault detected on the ground	Road-safe mode; thrust locked out until service	Warning; thrust functions greyed out
Gas below reserve before a hop	Lower manoeuvre offered	Message with the largest manoeuvre available
Fault during Boost	Thrust cut; car stays on its wheels	Warning
Fault during a hop	Recover with remaining gas: level first, then descend and land	Recovery indicator; harness pre-tension
Second fault during recovery	Backup law holds attitude and descends; external airbags armed	As above
Suction loss in Track mode	Speed and cornering limits within 100 ms	Warning

7.4 Technical safety requirements

Technical safety requirements say *how*, allocate each function to an architecture element of chapter 4 and name the verification item that will prove it. They are generated from the requirements database (`data/technical_safety_requirements.csv`).

ID	Technical safety requirement	ASIL	From	Allocated to	Verified by
TSR-01	Two normally closed master valves in series: valve A powered only by the safety monitor, valve B only by the thrust control unit	B(D) + B(D)	FSR-1.1	S4, S9, IF-01, IF-02	VER-01
TSR-02	Arming requires Track mode, a physical arm switch, zone authorisation from two independent sources, speed and wind within limits, belts fastened and a clear zone	D	FSR-1.2, FSR-7.1	S8, S9, S10, IF-05, IF-09	VER-02, VER-20
TSR-03	A leak-down test of the master valves runs at every power-on; a failure locks out thrust	D	FSR-1.1	S4, S8, S9	VER-01
TSR-04	Thrust messages are authenticated; production cars have no remote thrust path	D	FSR-1.3	S9, IF-03	VER-03, VER-21
TSR-05	Pressure is measured by two dissimilar sensors per gas group, with temperature compensation	D	FSR-6.4, FSR-2.1	S3, S8, IF-07	VER-04
TSR-06	COPVs meet a burst ratio of 2.25 or more and pass pressure-cycling, fire, drop and impact tests equivalent to UN GTR 13	Design standard	FSR-2.1	S3	VER-05
TSR-07	On crash detection all vessel valves close within 50 ms; venting only through thrust-neutral diffusers away from the cabin	D	FSR-2.2	S3, S4, S11, IF-10	VER-06
TSR-08	Loss of suction is detected from plenum pressure and speed and cornering limits apply within 100 ms, with a warning	C	FSR-4.1	S6, S7, IF-08	VER-07
TSR-09	A stuck-open thruster valve is detected and its line isolated within 20 ms (hard ceiling 50 ms)	B	FSR-5.1, FSR-6.3	S4, S5, S9, IF-11	VER-08

ID	Technical safety requirement	ASIL	From	Allocated to	Verified by
TSR-10	A hop starts only if usable gas is at least 1.2 times the predicted need, corrected for mass and gas temperature; otherwise a lower manoeuvre is offered	B	FSR-6.1	S3, S9	VER-09
TSR-11	If predicted gas at landing falls below the reserve mid-manoeuvre, recovery starts at once: level attitude first, then land	B	FSR-6.2	S9	VER-10
TSR-12	While airborne the safety monitor commands recovery instead of closing the master valves, unless thrust itself is the uncontainable fault	B	FSR-6.2	S9	VER-11
TSR-13	Attitude comes from three IMUs with voting; the monitor checks it with a dissimilar sensor	B	FSR-6.4	S8, IF-06	VER-12
TSR-14	External airbags arm only in Hop mode above 1.5 m and are disarmed by two independent signals on landing	B	FSR-8.1	S11	VER-13
TSR-15	After isolating any one lift group, the remaining torque-balanced groups lift at least 1.1 times maximum vehicle weight (about 10 kN per group)	B	FSR-6.3	S5	VER-14
TSR-16	Before every arming each thruster, line and master valve is cycled against a pressurised test volume; coverage of stuck-open and stuck-closed faults at least 99.9%	D	FSR-1.1, FSR-6.3	S4, S5, S9	VER-15
TSR-17	Any single stuck-closed nozzle leaves at least 50% of nominal roll-braking authority (individually controlled nozzles)	B	FSR-6.3	S5, S9	VER-16
TSR-18	COPVs, valves and seals are qualified to the lowest gas temperature of a full discharge, or discharge is limited to their qualified temperature	D	FSR-2.1	S3, S4, S5	VER-17
TSR-19	Refill air is dried to a dew point below -40 C, checked at the coupling; otherwise the fill is refused	D	FSR-2.1	S2	VER-18
TSR-20	The recovery planner chooses the upright attitude reachable with the least altitude loss, using the predicted stopping angle	B	FSR-6.2	S9	VER-19
TSR-21	In Hop mode no single loss of power, coil, switch or computer closes a master valve for at least 5 s (dual-wound coils, second driver, hold-up energy); closing in flight needs an explicit command	B	FSR-6.2	S1, S4, S9	VER-22
TSR-22	Hop-mode control is fail-operational for at least 5 s: a backup attitude-hold and descent law on independent hardware takes over within 50 ms	B	FSR-6.2	S9	VER-23
TSR-23	Thruster and line valve commands are direct-wired or carried on two independent buses; a lost command opens nothing new	B	FSR-6.2	S9, IF-03, IF-11	VER-24

8. Safety analyses

Five analyses check the architecture from different directions: the FMEA works up from component failures, the fault tree works down from the worst outcome, the FMEDA puts numbers on the hardware, the dependent failure analysis looks for faults that defeat both channels at once, and STPA looks for unsafe control decisions with nothing broken.

8.1 Subsystem FMEA

Subsystem	Failure mode	Worst effect	Main safety mechanism	Goal
S2 Compressor and fill	Over-pressure; wet air	Vessel overload; ice in valves	Fill limit plus mechanical relief; dew-point check	SG2
S3 Gas storage	Rupture; leak	Blast; loss of reserve in flight	Burst ratio, shielding; pressure decay monitor	SG2, SG6
S4 Distribution	Master or line valve stuck open or closed	Unintended thrust; loss of a lift group	Series masters; pre-arm test; line isolation	SG1, SG6
S5 Thrusters	Valve stuck open, stuck closed or slow	Asymmetric thrust	Line isolation and compensation	SG5, SG6
S6 Suction	Fan or skirt failure	Sudden loss of downforce	Plenum pressure monitoring, limits	SG4
S8 Sensing	IMU, pressure or positioning error	Wrong control or wrong zone	Voting, dissimilar checks	SG1, SG6
S9 Control	Wrong command; computer lost	Unintended thrust; loss of control	Two channels; backup hop law	SG1, SG6
S10 HMI and arming	Stuck arm switch; false request	Arming without consent	Two-step confirm plus physical switch	SG1
S11 Protection	Airbag fires on the road	Driver startled; loss of view	Arming window, two disarm signals	SG8

8.2 Component design FMEA

The component FMEA uses the AIAG-VDA method: severity (S), occurrence (O) and detection (D) are each rated 1–10 and combined into an action priority (H, M or L) from the AIAG-VDA table. Parts are representative types, not Tesla's parts (A-11). Of 31 failure modes, 23 are at action priority H. That is expected at concept stage, because most modes have S = 9 or 10 and design verification does not yet exist. The table below is generated from `results/component_fmea.csv`.

Component	Failure mode	Effect	S	O	D	AP	Goal	Field diagnostic	DC %
COPV (Type IV, ~100 L, 400 bar)	Rupture	Blast and fragments	10	3	2	L	SG2	Impact log, pressure-rate monitor	60
COPV (Type IV, ~100 L, 400 bar)	Liner or boss leak	Gas loss; reserve error in flight	9	4	3	H	SG6	Pressure decay vs model	95
Pressure relief device (burst disc + thermal)	Opens early	Uncontrolled vent; possible net thrust	8	4	3	M	SG2	Pressure drop	99
Pressure relief device (burst disc + thermal)	Fails to open in fire	Tank rupture in fire	10	4	2	H	SG2	None in service	0
Group isolation valve (NC solenoid)	Stuck open	Group cannot be isolated after crash	9	4	3	H	SG2	Power-on leak-down test	99

Component	Failure mode	Effect	S	O	D	AP	Goal	Field diagnostic	DC %
Group isolation valve (NC solenoid)	Stuck closed	Group gas unavailable; lower reserve	7	6	3	H	SG6	Pre-arm function test	99
Master valve A / B (NC solenoid, series)	Stuck open or seat leak	One SG1 barrier lost (latent)	10	4	3	H	SG1	Inter-stage and manifold pressure, power-on leak-down	99
Master valve A / B (NC solenoid, series)	Closes in flight (coil or power loss)	All thrust lost mid-hop	10	4	3	H	SG6	Coil current monitor	90
Pressure regulator (dome-loaded)	Fails high	Over-pressure in manifold; thrust excess	8	4	3	M	SG5	Manifold pressure	99
Pressure regulator (dome-loaded)	Fails low or freezes	Thrust shortfall in flight	10	4	3	H	SG6	Manifold pressure vs command	95
Line isolation valve (NC solenoid)	Stuck open	Stuck thruster cannot be isolated	10	4	3	H	SG6	Pre-arm function test	99
Line isolation valve (NC solenoid)	Stuck closed	Lift group lost	10	6	3	H	SG6	Pre-arm function test	99
Thruster valve (fast solenoid)	Stuck open	Asymmetric thrust; latent SG1 barrier loss	10	6	3	H	SG1/SG6	Line pressure vs command; pre-arm test	95
Thruster valve (fast solenoid)	Stuck closed	Roll authority lost (weak spot)	10	6	3	H	SG6	Pre-arm function test	95
Thruster valve (fast solenoid)	Slow response	Control degradation	7	6	4	H	SG6	Command vs pressure timing	90
Thruster valve (fast solenoid)	Seat leak	Small unwanted thrust; latent SG1	6	6	4	M	SG1	Power-on leak-down	90
Pressure transducer (strain gauge)	Drift or stuck in range	Wrong reserve estimate; hop starts short of gas	9	6	3	H	SG6	Cross-check with second sensor and gas model	99
Pressure transducer (strain gauge)	Open or short	Signal lost	6	6	3	M	SG6	Range check	99
IMU (MEMS)	Wrong output	Wrong attitude corrections	10	6	3	H	SG6	Voting	99
IMU (MEMS)	Loss of output	One attitude source lost	6	6	3	M	SG6	Timeout	99
Thrust control unit (lockstep MCU)	Wrong output	Erroneous thrust or valve command	10	4	2	H	SG1/SG6	Lockstep compare, monitor plausibility	99
Thrust control unit (lockstep MCU)	Loss of function in flight	No flight control until backup acts	10	4	3	H	SG6	Watchdog	99
Safety monitor (dissimilar MCU)	Wrong output (enable)	One SG1 barrier lost	10	4	2	H	SG1	Inter-stage pressure, TCU cross-check	90

Component	Failure mode	Effect	S	O	D	AP	Goal	Field diagnostic	DC %
Safety monitor (dissimilar MCU)	Loss of function in flight	Master valve A hold lost	10	4	3	H	SG6	TCU watchdog of monitor	99
High-side switch, master valve A / B	Short (stuck on)	One SG1 barrier lost	10	4	3	H	SG1	Output read-back	99
High-side switch, master valve A / B	Open	Master valve closes in flight	10	4	3	H	SG6	Coil current	99
Low-side valve driver	Short (valve on)	Behaves as stuck-open valve	10	4	3	H	SG1/SG6	Driver diagnostics	95
Low-side valve driver	Open (valve off)	Behaves as stuck-closed valve	10	4	3	H	SG6	Open-load detection	95
Arm-line harness, each channel	Short to battery	One SG1 barrier lost	10	3	4	L	SG1	Read-back	90
Satellite and beacon receivers	False position or replayed beacon	Zone wrongly authorised (needs both)	8	4	3	M	SG1	Cross-check with inertial data	90
Low-voltage supply, thrust domain	Loss in flight	Valves close; thrust lost	10	4	3	H	SG6	Voltage monitor	99

Findings that changed the architecture.

1. *Master valve closes in flight (coil or power loss).* A normally closed valve is the right choice on the ground but drops the car in the air. This led to TSR-21 and AD-04.
2. *Thrust control unit lost in flight.* With no backup control law, the monitor can only cut gas, which Sim 2 shows is fatal. This led to TSR-22.
3. *Valve command bus lost.* A shared bus is a single point for every nozzle. This led to TSR-23.
4. *Pressure regulator freezes.* A full discharge chills the gas to about 142 K. This is covered by TSR-18 and TSR-19.

8.3 FMEDA and hardware metrics for SG1

The fault tree for SG1 (no unintended thrust) was solved for minimal cut sets, and each cut set was assigned a rate from the component failure rates, diagnostic coverage and pre-arm test interval: $q = \lambda \cdot [DC \cdot \tau/2 + (1 - DC) \cdot T/2]$, with τ = 1 h between pre-arm tests and T = 10,000 h vehicle life (A-12, A-13).

Metric	Result	ASIL D target	Met
Safety-related failure rate	1,236 FIT	—	—
Single-point fault metric (SPFM)	100.0% (rounded; only the computer common-cause term is single-point)	≥ 99%	Yes
Latent fault metric (LFM)	98.7%	≥ 90%	Yes
Probabilistic metric for hardware failures (PMHF)	0.006 FIT	< 10 FIT	Yes
Minimal cut sets	36: one of order 1, two of order 2, 24 of order 3, nine of order 4	—	—

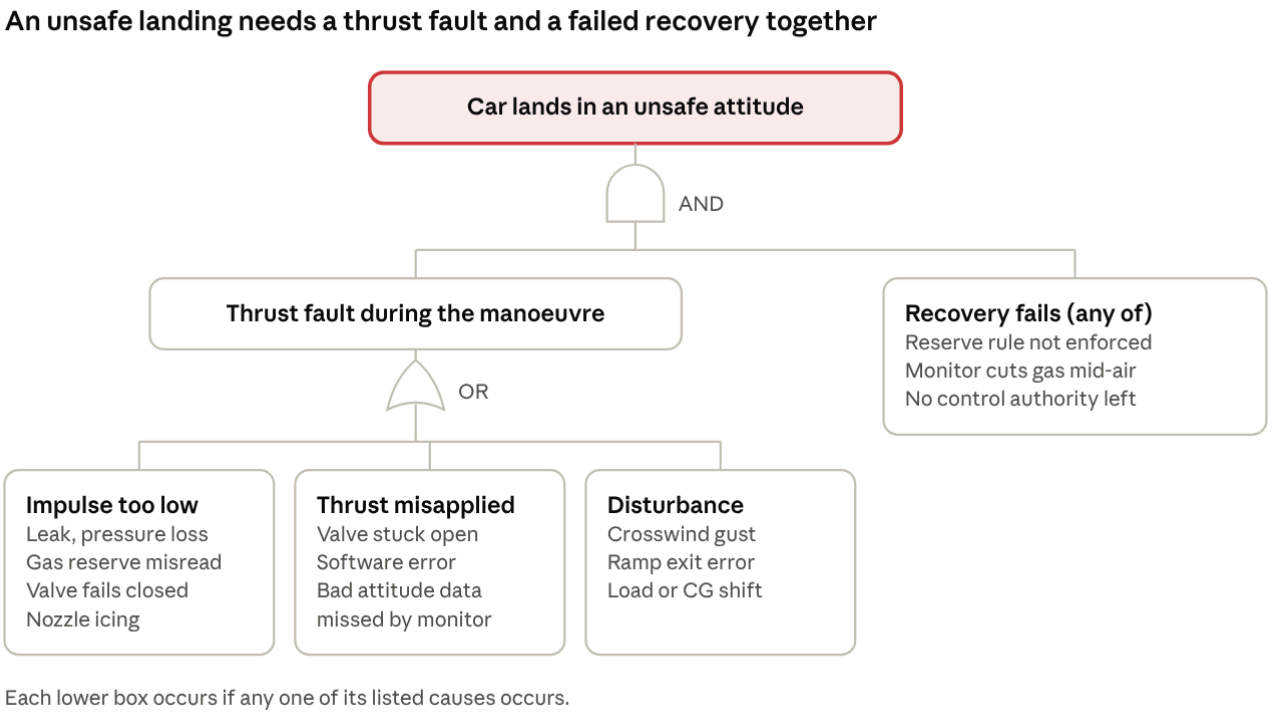


Figure 5. Fault tree for an unsafe landing after a hop (SG6): a thrust fault (too little impulse, misapplied thrust or a disturbance) must combine with a failed recovery.

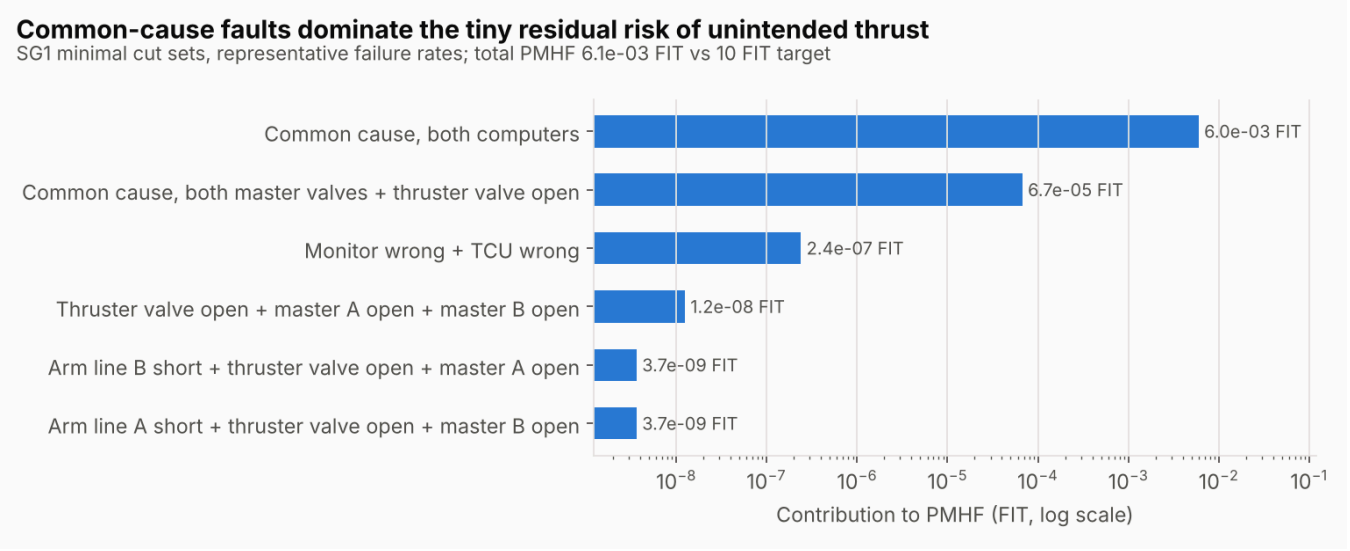


Figure 6. The largest SG1 cut sets by rate. Common-cause failure of the two computers dominates; every independent combination is at least three orders of magnitude smaller.

How to read these numbers. The metrics pass by a wide margin, but they rest on assumed failure rates and β-factors (1% for the computers and 2% for the master valves, A-14). The ranking matters more than the absolute values: the residual is dominated by common-cause faults, so the dependent failure analysis is the decisive evidence for SG1. An earlier, coarser FMEDA (Sim 6) gave PMHF 0.053 FIT; it is superseded by this component-level result.

8.4 Fault tree for SG6

An unsafe landing needs a thrust fault and a failed recovery together (Figure 5). That AND gate is the reason recovery logic matters so much. Per-hop probabilities from Sim 6, with 11 kN lift groups and a 99.9%-coverage pre-arm valve test, are:

Branch	Probability per hop
Stuck-closed thruster valve, latent at take-off, during the roll	1.8×10^{-7}
Leak	1.4×10^{-7}
Stuck-open thruster valve	1.4×10^{-8}
Total from random hardware faults	**3.3×10^{-7}** (target 10^{-7}, not yet met)

Stuck-closed nozzles during the roll dominate. Meeting the target needs either individually controlled nozzles that keep at least 50% roll-braking authority (TSR-17) or a better recovery planner (TSR-20). Both show gaps in simulation today.

8.5 Dependent failure analysis

Coupling factor	Could defeat	Measure	Residual
Same microcontroller design or compiler	Both computers	Dissimilar monitor, different supplier and tool chain (AD-02)	β about 1%
Same low-voltage supply	Both computers; both master valves	Separate feeds; monitor's own buffer; hold-up store (TSR-21)	Low
Same zone data	Both channels' arming decisions	Independent zone check in the monitor (CC-13, AD-09)	Low after CC-13
Same valve type and batch	Both master valves	Different valve suppliers or seat designs; leak-down test (TSR-03)	β about 2%
Cold from gas expansion	Valves, seals, regulator	Cold qualification (TSR-18); dry air (TSR-19)	Needs test
Ice or contamination in the gas	All valves	Filter and dew-point check (TSR-19)	Low
Vessel burst fragments	Harness, both computers	Segregation, shielding and routing rules	Needs layout
Vibration and shock of landing	Connectors, sensors	Qualification to landing loads (Sim 5)	Needs test
Same software requirement error	Both channels	Different teams and requirements derived independently from FSRs	Process measure
Cyber attack on a shared input	Both channels	Isolated domain, authenticated bus (chapter 10)	See TARA

8.6 STPA unsafe control actions

STPA covers hazards that need no failure: the controller does exactly what it was designed to do, but the decision is wrong for the situation.

Control action	Not provided	Provided when unsafe	Wrong timing or order	Stopped too soon or applied too long
Arm thrust	—	UCA-1: arming with a person in the clear zone (SG7)	UCA-2: arming before the pre-arm test finishes (SG1)	—
Open master valves	UCA-3: master valves not held open mid-hop (SG6)	UCA-4: open on a public road (SG1)	—	UCA-5: closed mid-hop by a supply dip (SG6)
Fire thrusters for a hop	—	UCA-6: hop with too little gas for recovery (SG6)	UCA-7: roll starts before climb rate settles (SG6)	UCA-8: thrust held after touchdown, car slides (SG7)

Control action	Not provided	Provided when unsafe	Wrong timing or order	Stopped too soon or applied too long
Command recovery	UCA-9: fault detected, no recovery commanded (SG6)	UCA-10: recovery on a false alarm wastes gas (SG6)	UCA-11: recovery after the FTTI has passed (SG6)	UCA-12: recovery ends before the car is level (SG6)
Cut all gas	—	UCA-13: cut while airborne (SG6)	—	—
Deploy external airbags	UCA-14: no deployment on a hard landing (SG8)	UCA-15: deployment on the road (SG8)	UCA-16: deployment too early, airbag vents before contact (SG8)	—

Every unsafe control action maps to an existing requirement: UCA-1 to TSR-02; UCA-2 to TSR-16; UCA-3, UCA-5 and UCA-13 to TSR-12 and TSR-21; UCA-4 to TSR-01 and TSR-02; UCA-6 to TSR-10; UCA-7 to TSR-11 and TSR-20; UCA-8 to a landing-detection requirement to be added in the next issue (open issue OI-07); UCA-9 to UCA-12 to TSR-09, TSR-11, TSR-12 and TSR-20; UCA-14 to UCA-16 to TSR-14.

9. SOTIF analysis

ISO 21448 deals with hazards that arise when nothing is broken: the sensors, models or control laws are simply not good enough for the situation. For a hop these are the main risk, because the physics is unforgiving and the domain is new.

9.1 Triggering conditions

ID	Triggering condition	Functional insufficiency	Possible hazard	Measure
TC-01	Gust above the planned limit during the roll	Control authority too low	Inverted landing	Wind limit in the operating domain (A-15); gust term in the reserve check
TC-02	Gas colder than predicted after a fast refill	Reserve over-estimated	Gas runs out before landing	Temperature-corrected reserve (TSR-10)
TC-03	Mass or centre of gravity outside the assumed range (luggage, passenger)	Controller tuned for wrong inertia	Roll overshoot	Seat and load sensing; refuse hop if out of range
TC-04	Ground effect and jet reflection near the surface	Thrust model wrong below about 1 m	Bounce or tilt at take-off and landing	Ground-effect term; thrust-stand data
TC-05	Loose gravel or debris	Jets throw debris; suction skirt lifts	Bystander injury; tilt	Surface check in the operating domain
TC-06	Satellite multipath at the track	Position error	Arming outside the zone	Beacon plus inertial cross-check (CC-13)
TC-07	Vibration or ice affects pressure sensing	Biased gas estimate	Wrong reserve decision	Dual dissimilar sensors (TSR-05)
TC-08	Driver input during the hop	Mode confusion	Unexpected manoeuvre	Driver inputs ignored in Hop mode except abort
TC-09	Nozzle thrust scatter above the 3% (1σ) modelled in Sim 4	Allocation error	Drift and roll error	Closed-loop thrust estimation; thrust-stand calibration
TC-10	Landing surface slope or step	Landing law assumes level ground	Roll on touchdown	Landing pad specification

9.2 Known unsafe scenarios found so far

Sim 4 found no bad outcomes in 2,186 fault-free hops under random fill, temperature, mass, centre-of-gravity, gust and thrust scatter once the reserve check was in place. Without the reserve check, 21% of fault-free hops ran out of gas: that was the largest SOTIF finding of this study and led directly to TSR-10.

9.3 Acceptance criterion and validation target

Item	Value
Acceptance criterion	Fewer than 10^{-7} catastrophic outcomes per hop from SOTIF causes, the same as for random hardware faults
Evidence today	Zero bad outcomes in 2,186 fault-free simulated hops gives an upper bound of 1.4×10^{-3} per hop at 95% confidence
Validation target	About 3×10^{7} hop-equivalents with no catastrophic outcome (zero-event rule: $n \geq 3 / 10^{-7}$)
How to get there	Mostly simulation with importance sampling of the triggering conditions above, anchored by about 10^{3} instrumented physical hops that validate the models

9.4 Validation strategy

1. **Model validation.** Thrust-stand tests of each nozzle and of nozzle groups near the ground; tank blowdown tests at cold and hot fills; drop tests for the landing model.
2. **Scenario-based simulation.** A 6-degree-of-freedom model, validated in step 1, run over the triggering-condition space with importance sampling at the edges (gust in the roll, cold gas, mass offset).
3. **HIL.** The real thrust control unit and safety monitor in closed loop with the validated model.
4. **Tethered hops.** Physical hops on a crane tether with an empty cabin, then with ballast, then with a test pilot.
5. **Field monitoring.** Every hop logged; any recovery, refusal or limit exceedance reviewed.

10. Cybersecurity concept

A thrust system is a cybersecurity problem with physical consequences: a successful attack on it can fire jets on a public road. The analysis follows ISO/SAE 21434 and is designed to support a UN R155 type approval and a UN R156 software-update management system.

10.1 Assets

ID	Asset	Property to protect
A1	Thrust commands and valve states	Integrity, availability in flight
A2	Mode and arming state	Integrity
A3	Zone authorisation (satellite fix, beacon)	Integrity, authenticity
A4	Thrust-domain software and calibration	Integrity, authenticity
A5	Cryptographic keys	Confidentiality, integrity
A6	Gas state and fill commands	Integrity
A7	Flight and location logs	Confidentiality
A8	Update packages	Integrity, authenticity
A9	Diagnostic and service access	Integrity, access control

10.2 Security architecture

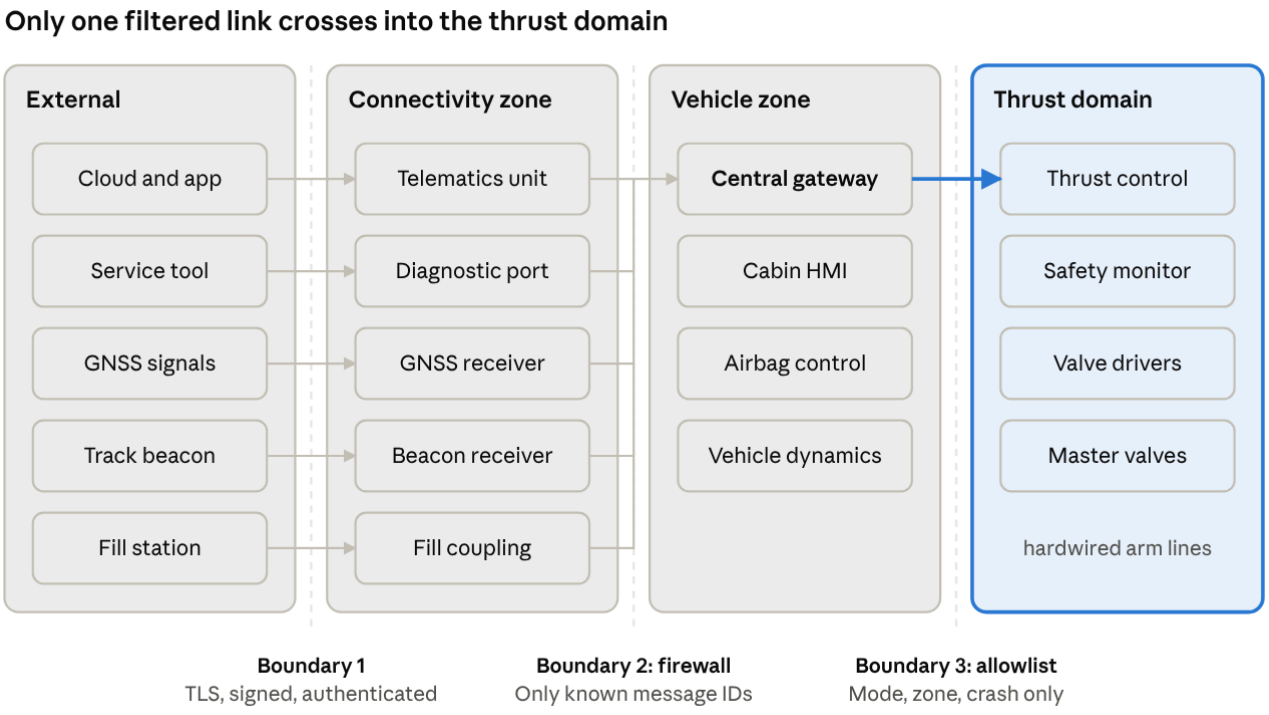


Figure 7. Security zones: the thrust domain is isolated behind a zonal gateway that passes only three signal types; connectivity and infotainment have no path to it.

10.3 Damage scenarios

ID	Damage scenario	Impact	Linked safety goal
DS1	Thrusters fire on a public road or near people	Severe	SG1

ID	Damage scenario	Impact	Linked safety goal
DS2	Loss of control during a hop	Severe	SG6
DS3	Thrust enabled outside an authorised zone	Severe	SG1, SG7
DS4	COPV over-pressure or rupture while filling	Severe	SG2
DS5	Fleet-wide unsafe or disabling software	Severe	SG1, SG6
DS6	Vehicle held to ransom; thrust unavailable	Major	-
DS7	Owner removes safety limits	Severe	SG1, SG6, SG7
DS8	Location and flight logs exposed	Major	-

10.4 Threat analysis and risk assessment

Attack feasibility uses the attack-potential method of ISO/SAE 21434 Annex G. Five factors are scored: elapsed time, specialist expertise, knowledge of the item, window of opportunity and equipment. A total of 0–13 points is High feasibility, 14–19 Medium, 20–24 Low and 25 or more Very low. Risk value (1–5) comes from impact and feasibility. Residual risk above 3 is not accepted, and the traceability check enforces this rule.

ID	Threat scenario	Damage	Impact	Factors before	Feasibility before	Risk before	Controls	Factors after	Feasibility after	Risk after
TS1	Remote pivot from telematics to thrust command	DS1	Severe	4-6-3-0-4 = 17	Medium	4	CC-01, CC-02	19-8-7-0-7 = 41	Very low	2
TS2	Diagnostic-port injection of mode and zone data	DS1, DS3	Severe	1-3-0-1-0 = 5	High	5	CC-01, CC-03	4-6-3-4-4 = 21	Low	3
TS3	Satellite spoofing plus beacon replay	DS3	Severe	1-3-0-1-4 = 9	High	5	CC-04, CC-05, CC-13	4-6-3-4-4 = 21	Low	3
TS4	Owner reflashes thrust software	DS7	Severe	1-3-0-0-4 = 8	High	5	CC-06, CC-07	4-8-7-0-4 = 23	Low	3
TS5	Malicious or tampered software update	DS1, DS5	Severe	4-6-7-4-0 = 21	Low	3	CC-06, CC-08	19-8-11-10-7 = 55	Very low	2
TS6	Spoofed fill station commands over-pressure	DS4	Severe	4-3-3-1-4 = 15	Medium	4	CC-09	19-6-7-4-7 = 43	Very low	2
TS7	Bus jamming during a hop	DS2	Severe	0-3-0-4-0 = 7	High	5	CC-02, CC-10	4-6-3-4-4 = 21	Low	3
TS8	Theft of flight logs from the cloud	DS8	Major	4-6-3-0-4 = 17	Medium	3	CC-11	4-6-7-0-4 = 21	Low	2
TS9	Ransomware through infotainment	DS6	Major	4-6-3-0-4 = 17	Medium	3	CC-01, CC-12	4-6-7-0-4 = 21	Low	2

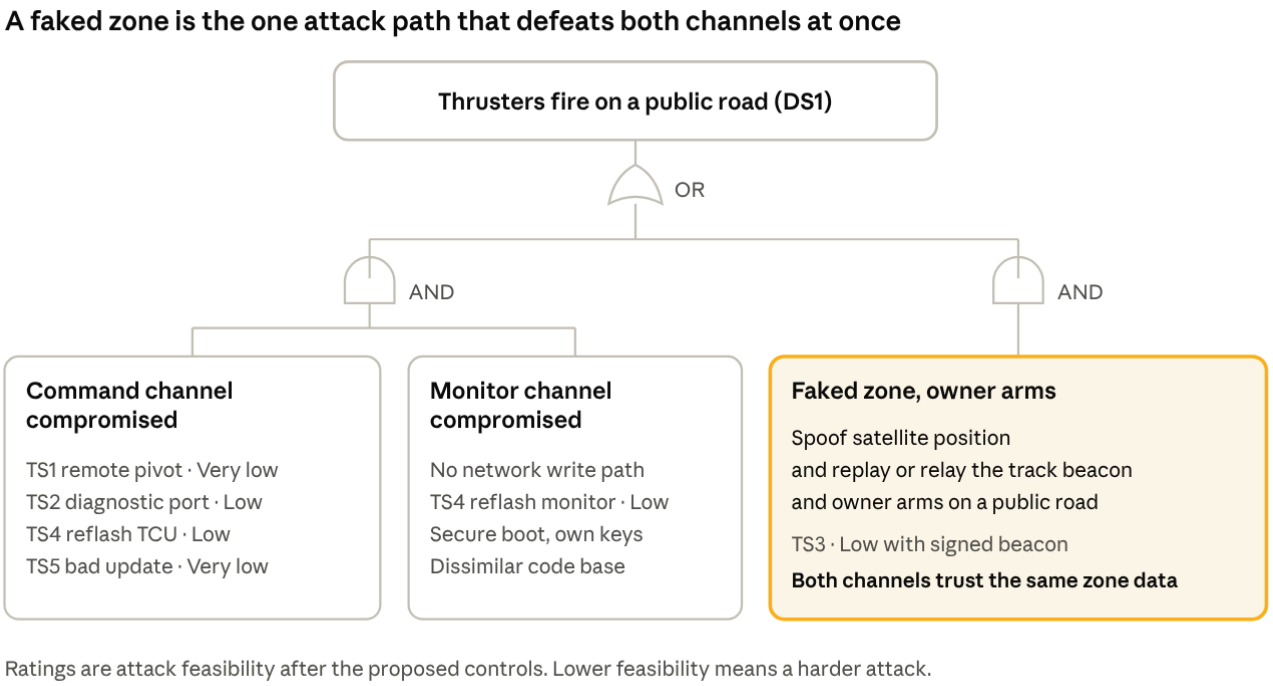


Figure 8. Attack tree for "thrusters fire on a public road". A faked track zone is the one path that defeats both channels at once, because both trust the same zone data; the independent zone check (CC-13) closes it.

10.5 Cybersecurity goals

ID	Cybersecurity goal	CAL	Threats
CG1	No thrust command or mode change can enter the thrust domain from outside it	CAL 4	TS1, TS2
CG2	Zone authorisation cannot be faked by spoofing or replay	CAL 4	TS3
CG3	Only genuine, approved software runs in the thrust domain	CAL 4	TS4, TS5
CG4	No external party can command a fill above the rated pressure	CAL 3	TS6
CG5	Loss or jamming of thrust-domain communication never leads to loss of control	CAL 3	TS7
CG6	Personal data and non-safety functions are protected and segregated	CAL 2	TS8, TS9

10.6 Cybersecurity controls

ID	Control	Goal	Verified by
CC-01	Zonal gateway with allowlist: only mode request, zone data and crash signal reach the thrust domain	CG1, CG6	VER-03
CC-02	Private thrust bus with message authentication, counters and timeouts	CG1, CG5	VER-21
CC-03	Diagnostic access needs an authenticated service tool and empty tanks	CG1	VER-21
CC-04	Signed challenge-response track beacon with distance bounding	CG2	VER-25
CC-05	Positioning plausibility: satellite fix checked against wheel odometry and inertial data	CG2	VER-25
CC-06	Secure boot with a hardware root of trust in both thrust computers	CG3	VER-21
CC-07	Tamper detection forces Road-safe mode and logs the event	CG3	VER-21
CC-08	Signed update packages, anti-rollback; thrust-domain updates only with tanks empty and the car present	CG3	VER-21

ID	Control	Goal	Verified by
CC-09	Authenticated fill coupling; independent mechanical pressure relief	CG4	VER-18
CC-10	Communication loss in flight triggers the backup hop law (TSR-22)	CG5	VER-23
CC-11	Log encryption, data minimisation and retention limits	CG6	VER-21
CC-12	Domain segmentation: infotainment compromise cannot reach chassis or thrust	CG6	VER-03
CC-13	Safety monitor confirms the zone independently by inertial dead-reckoning from the last trusted beacon fix	CG2	VER-25

10.7 Operations and post-development

Activity	Practice	Regulation
Cybersecurity management system	Organisation-level processes, roles and audits	UN R155
Software update management	Signed updates; thrust domain updated only with tanks empty and the car present; version records per vehicle	UN R156
Vulnerability monitoring	Supplier bulletins, bug bounty, fleet security monitoring	ISO/SAE 21434 cl. 8
Incident response	Fleet-wide remote lockout of thrust functions (to Road-safe only, never to enable)	ISO/SAE 21434 cl. 13
End of cybersecurity support	Thrust functions disabled when support ends	ISO/SAE 21434 cl. 14

The remote lockout can only *disable* thrust. No remote command can enable it, which keeps the incident-response path from becoming an attack path.

11. Verification and validation plan

11.1 Strategy

Verification asks "did we build it right?" against each requirement. Validation asks "did we build the right thing?" against the safety goals and the acceptance criteria. Both follow the V-model, from analysis at concept stage to vehicle tests.

Level	Methods	Main questions
Concept	FMEA, FMEDA, FTA, DFA, STPA, TARA, simulation	Is the architecture capable of meeting the goals?
Component	Valve, sensor and vessel qualification; burst, cycle, cold and fire tests	Do parts meet their failure-rate and environmental assumptions?
Software	Requirements-based tests, MC/DC coverage for ASIL D parts, static analysis, fuzzing	Does the code do what the requirements say, and nothing else?
Integration (HIL)	Real computers in closed loop with a validated model; fault and attack injection	Do detection and reaction meet the timing budget?
Vehicle	Thrust stand, tethered hops, drop tests, crash tests with charged tanks, track tests	Does the whole car behave as predicted?
Validation	Scenario-based simulation at scale, field monitoring	Is residual risk below 10^{-7} per hop?

11.2 Verification items

Every technical safety requirement and every cybersecurity control has at least one verification item, and every verification item lists what it covers. The table is generated from `data/verification.csv`.

ID	Covers	Method	Pass criterion	Status
VER-01	TSR-01, TSR-03	FMEDA; HIL fault injection; vehicle leak-down test	SPFM ≥ 99%, LFM ≥ 90%, PMHF < 10 FIT; no gas flow after any single fault	Analysis done
VER-02	TSR-02	HIL test of every combination of arming inputs	Arms only when all conditions hold	Not started
VER-03	TSR-04, CC-01, CC-12	Architecture review; penetration test	No external route to a thrust command	Not started
VER-04	TSR-05	HIL sensor-fault injection	Any single sensor fault detected	Not started
VER-05	TSR-06	Burst, cycling, fire, drop and impact tests	Burst ratio ≥ 2.25; all tests passed	Not started
VER-06	TSR-07	Valve timing bench; crash test with charged tanks	Valves closed ≤ 50 ms; no projectile; no net vent thrust	Not started
VER-07	TSR-08	Track test with controlled skirt leak; HIL	Limits applied ≤ 100 ms; car stays on line	Not started
VER-08	TSR-09	Sim 3; HIL timing measurement	≤ 20 ms measured; FTTI ≥ 200 ms confirmed	Simulated
VER-09	TSR-10	Sim 4; track test with reduced fill	No gas-starved landing	Simulated
VER-10	TSR-11	Sim 4; HIL	Leak cases land safely	Simulated
VER-11	TSR-12	Sim 2; HIL	Upright landing after a mid-roll fault	Simulated
VER-12	TSR-13	HIL sensor-fault injection	Single IMU fault masked	Not started
VER-13	TSR-14	Sim 5; instrumented drop tests	Peak ≤ 15 g from 5 m; no deployment below 1.5 m or on road	Simulated
VER-14	TSR-15	Sim 3, Sim 4; thrust stand	Remaining lift ≥ 1.1 x weight	Simulated
VER-15	TSR-16	Diagnostic coverage analysis; HIL	Coverage ≥ 99.9% for stuck-open and stuck-closed	Analysis done

ID	Covers	Method	Pass criterion	Status
VER-16	TSR-17	Sim 4; design study	≥ 50% roll-braking authority remains	Gap found
VER-17	TSR-18	Cold-soak and full-discharge tests of COPVs and valves	No leak, crack or sticking at minimum temperature	Not started
VER-18	TSR-19, CC-09	Fill-station test	Wet air refused	Not started
VER-19	TSR-20	Sim 3 sweep; HIL	No bad outcomes at delays ≤ 50 ms	Gap found
VER-20	TSR-02	Track test with dummy targets in the clear zone	Arming refused with a target in the zone	Not started
VER-21	TSR-04, CC-02, CC-03, CC-06, CC-07, CC-08, CC-11	Fuzzing, penetration test, code review (CG1-CG3)	No successful attack at CAL 4 depth	Not started
VER-22	TSR-21	HIL injection of power, coil, switch and computer loss in Hop mode	No master valve closes within 5 s of any single fault	Analysis done
VER-23	TSR-22, CC-10	HIL controller-loss injection; extended Sim 2	Backup takes over ≤ 50 ms; upright landing ≤ 5 m/s	Analysis done
VER-24	TSR-23	HIL bus-loss injection	No valve opens unintentionally; attitude held	Analysis done
VER-25	CC-04, CC-05, CC-13	Spoofing and replay attack on a test track; HIL with simulated satellite and beacon signals	No arming with a spoofed fix or replayed beacon; monitor flags a zone mismatch within 1 s	Not started

11.3 Test scenarios

ID	Scenario	Fault or condition	Pass criterion	Covers
SC-01	Public-road drive with a forged arm request	Arm request injected on the vehicle bus	No master valve opens	TSR-01, TSR-02, TSR-04
SC-02	Power-on with a leaking master valve	Seat leak on valve A	Thrust locked out	TSR-03
SC-03	Stuck-open thruster at the apex	One nozzle forced open	Line isolated ≤ 20 ms; upright landing	TSR-09, TSR-15
SC-04	Stuck-closed nozzle during the roll	One nozzle forced closed at 1.3 s	Upright landing at ≤ 5 m/s	TSR-17, TSR-20
SC-05	Leak from one vessel group mid-climb	5 kg/s leak	Recovery starts at once; upright landing	TSR-10, TSR-11
SC-06	Thrust control unit reset mid-hop	Controller power removed	Backup law takes over ≤ 50 ms	TSR-22
SC-07	Supply dip mid-hop	Low-voltage feed drops for 200 ms	No master valve closes	TSR-21
SC-08	Valve bus lost mid-hop	Bus disconnected	No valve opens unintentionally; attitude held	TSR-23
SC-09	Crash with charged tanks	Frontal crash test	Vessels isolated ≤ 50 ms; no projectile; no net vent thrust	TSR-06, TSR-07
SC-10	Satellite spoofing at the track gate	Fake fix inside the zone	Arming refused; monitor flags mismatch	CC-04, CC-05, CC-13
SC-11	Hard landing from 5 m	Drop test with external airbag	Peak ≤ 15 g	TSR-14
SC-12	Cold refill then immediate hop	Gas 20 K colder than nominal	Reserve check corrects for temperature	TSR-10, TSR-18

11.4 Simulation credibility

The simulations in this study are concept-stage evidence. Before they can support a release decision they need the following.

Credibility factor	Status today	Needed
Model fidelity	Planar roll-plane rigid body; lumped tank	6-degree-of-freedom model with nozzle plume and ground effect
Parameter basis	Assumptions A-01 to A-16	Measured mass properties, thrust-stand and blowdown data
Code verification	Energy and momentum checks; deterministic seeds	Independent review; unit tests against analytic cases
Validation	None against hardware	Correlation with tethered hops within agreed tolerances
Tool qualification	Open-source Python	Tool confidence level assessed per ISO 26262-8 cl. 11

11.5 Roadmap

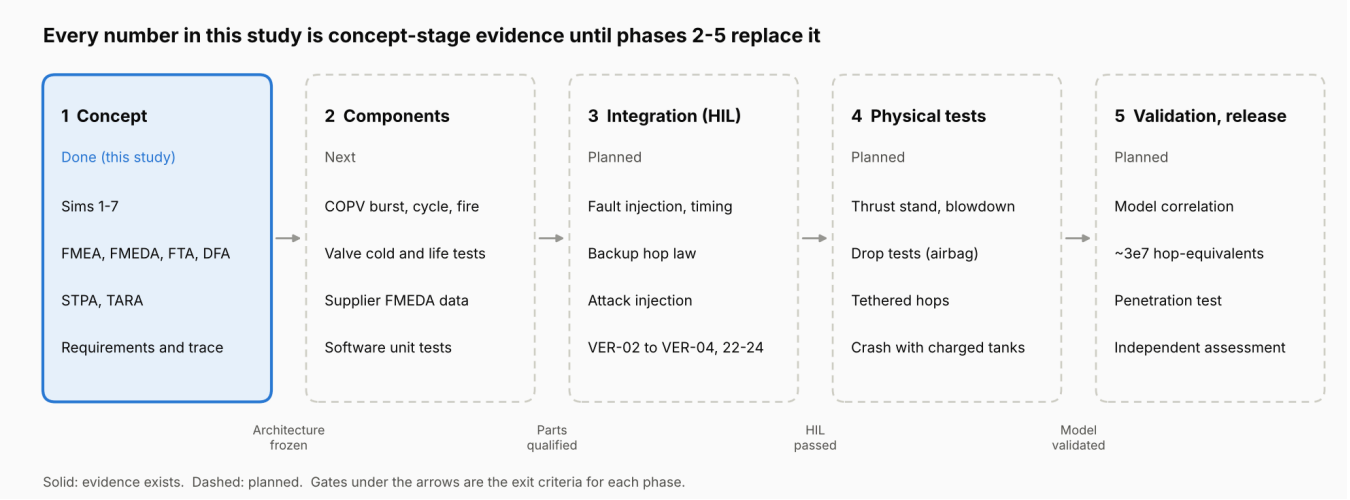


Figure 9. Verification and validation roadmap from this concept study (Sims 1–7) to release.

11.6 Evidence status

Of 25 verification items, 5 have analysis evidence, 6 have simulation evidence, 2 show a gap that needs design work (VER-16, VER-19) and 12 have not started. No physical test evidence exists.

12. Traceability

Traceability proves that every hazard is covered by a goal, every goal by requirements, every requirement by an architecture element and a test, and every test by a requirement. Two chains are traced.

- **Safety:** hazardous event → safety goal → functional safety requirement → technical safety requirement → architecture element → verification item → evidence.
- **Security:** damage scenario → threat scenario → cybersecurity goal → control → verification item.

All links live in one place, the requirements database (`tools/requirements_db.py`, written out to `data/*.csv`). The tables in this document are generated from it, so the document and the data cannot drift apart. `tools/check_traceability.py` checks every link in both directions and fails the build if any link is broken.

12.1 Safety trace matrix

TSR	Hazards	Goals	FSR	Elements	Verification	Status
TSR-01	HE1, HE2	SG1	FSR-1.1	S4, S9, IF-01, IF-02	VER-01	Analysis done
TSR-02	HE1, HE2, HE9	SG1, SG7	FSR-1.2, FSR-7.1	S8, S9, S10, IF-05, IF-09	VER-02, VER-20	Not started
TSR-03	HE1, HE2	SG1	FSR-1.1	S4, S8, S9	VER-01	Analysis done
TSR-04	HE1, HE2	SG1	FSR-1.3	S9, IF-03	VER-03, VER-21	Not started
TSR-05	HE3, HE7, HE8, HE11	SG2, SG6	FSR-6.4, FSR-2.1	S3, S8, IF-07	VER-04	Not started
TSR-06	HE3	SG2	FSR-2.1	S3	VER-05	Not started
TSR-07	HE3	SG2	FSR-2.2	S3, S4, S11, IF-10	VER-06	Not started
TSR-08	HE5	SG4	FSR-4.1	S6, S7, IF-08	VER-07	Not started
TSR-09	HE6, HE7, HE8, HE11	SG5, SG6	FSR-5.1, FSR-6.3	S4, S5, S9, IF-11	VER-08	Simulated
TSR-10	HE7, HE8, HE11	SG6	FSR-6.1	S3, S9	VER-09	Simulated
TSR-11	HE7, HE8, HE11	SG6	FSR-6.2	S9	VER-10	Simulated
TSR-12	HE7, HE8, HE11	SG6	FSR-6.2	S9	VER-11	Simulated
TSR-13	HE7, HE8, HE11	SG6	FSR-6.4	S8, IF-06	VER-12	Not started
TSR-14	HE10	SG8	FSR-8.1	S11	VER-13	Simulated
TSR-15	HE7, HE8, HE11	SG6	FSR-6.3	S5	VER-14	Simulated
TSR-16	HE1, HE2, HE7, HE8, HE11	SG1, SG6	FSR-1.1, FSR-6.3	S4, S5, S9	VER-15	Analysis done
TSR-17	HE7, HE8, HE11	SG6	FSR-6.3	S5, S9	VER-16	Gap found
TSR-18	HE3	SG2	FSR-2.1	S3, S4, S5	VER-17	Not started
TSR-19	HE3	SG2	FSR-2.1	S2	VER-18	Not started
TSR-20	HE7, HE8, HE11	SG6	FSR-6.2	S9	VER-19	Gap found
TSR-21	HE7, HE8, HE11	SG6	FSR-6.2	S1, S4, S9	VER-22	Analysis done
TSR-22	HE7, HE8, HE11	SG6	FSR-6.2	S9	VER-23	Analysis done
TSR-23	HE7, HE8, HE11	SG6	FSR-6.2	S9, IF-03, IF-11	VER-24	Analysis done

12.2 Security trace matrix

Damage	Threat	Goal	Controls	Verification	Risk
DS1	TS1	CG1 (CAL 4)	CC-01, CC-02	VER-03, VER-21	4 → 2
DS1, DS3	TS2	CG1 (CAL 4)	CC-01, CC-03	VER-03, VER-21	5 → 3
DS3	TS3	CG2 (CAL 4)	CC-04, CC-05, CC-13	VER-25	5 → 3

Damage	Threat	Goal	Controls	Verification	Risk
DS7	TS4	CG3 (CAL 4)	CC-06, CC-07	VER-21	5 → 3
DS1, DS5	TS5	CG3 (CAL 4)	CC-06, CC-08	VER-21	3 → 2
DS4	TS6	CG4 (CAL 3)	CC-09	VER-18	4 → 2
DS2	TS7	CG5 (CAL 3)	CC-02, CC-10	VER-21, VER-23	5 → 3
DS8	TS8	CG6 (CAL 2)	CC-11	VER-21	3 → 2
DS6	TS9	CG6 (CAL 2)	CC-01, CC-12	VER-03	3 → 2

12.3 Automated check

The output below was produced by the build that made this document.

```
hazards 11, goals 8, FSR 14, TSR 23, elements 22, verification items 25
damage scenarios 8, threats 9, cybersecurity goals 6, controls 13
verification status: Analysis done 5, Gap found 2, Not started 12, Simulated 6
out of scope: FSR-3.1
traceability check passed: every link resolves
```

Rules the check enforces.

1. Every hazardous event names a safety goal that exists, and every safety goal has at least one hazardous event.
2. Every safety goal has at least one functional safety requirement.
3. Every functional safety requirement is refined by a technical safety requirement, or is explicitly allocated to another item (FSR-3.1, steering).
4. Every technical safety requirement is allocated to an architecture element that exists, and names a verification item that exists and lists it back.
5. Every threat scenario names damage scenarios, a cybersecurity goal and controls that exist, and has residual risk 3 or below.
6. Every cybersecurity goal has a threat and a control; every control treats a threat and names a verification item that lists it back.
7. Every verification item covers only requirements or controls that exist.

13. Safety case, regulation, review record and open issues

13.1 Safety case structure

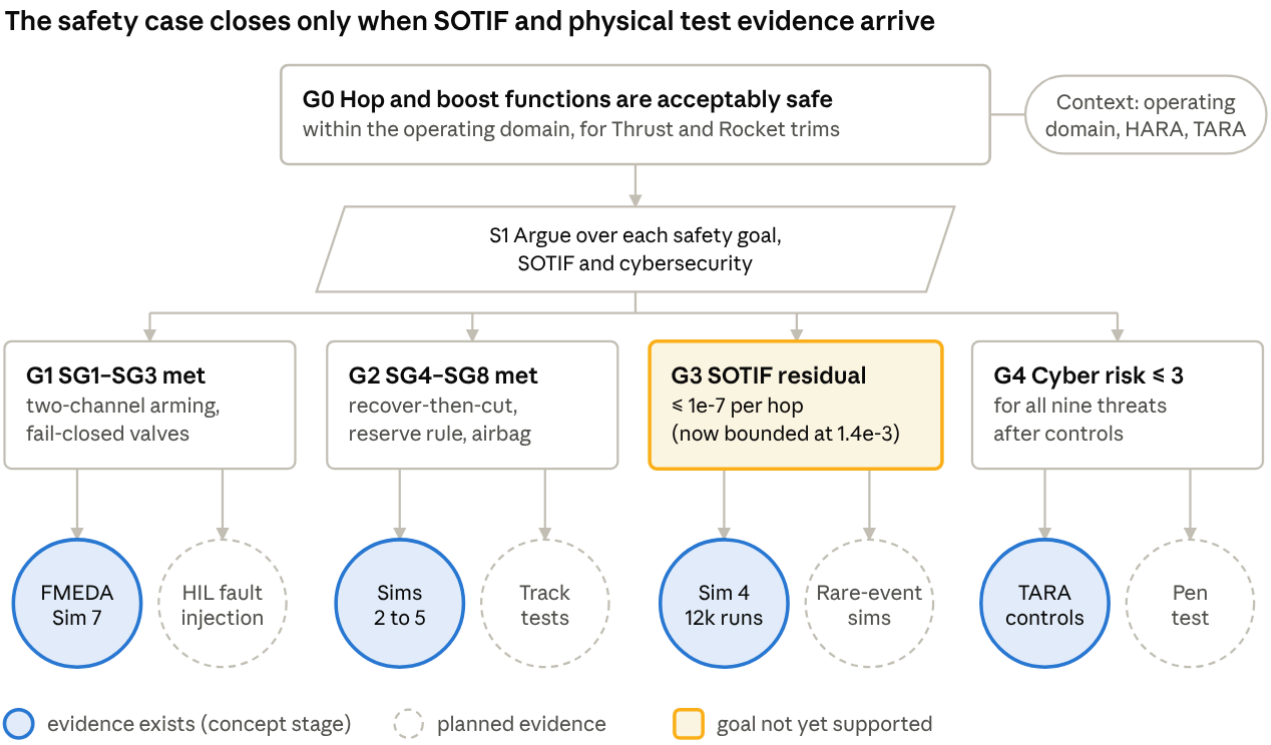


Figure 10. Safety case in goal structuring notation. The top claim (G0) is that the hop and boost functions are acceptably safe within the operating domain. Strategy S1 argues over each safety goal, SOTIF and cybersecurity.

Claim	Status	Evidence today	Evidence still needed
G1 SG1–SG3 are met	Supported at concept stage	FMEDA and cut sets (Sim 7): SPFM 100%, LFM 98.7%, PMHF 0.006 FIT	HIL fault injection; DFA with test data; steering item's own case (SG3)
G2 SG4–SG8 are met	Partly supported	Sims 2–5: recover-then-cut, reserve check, airbag loads	SG6 per-hop risk 3.3 × 10^{-7} against a 10^{-7} target; track tests
G3 SOTIF residual risk is at most 10^{-7} per hop	Not supported	Sim 4 bounds it at 1.4 × 10^{-3} per hop	Validated model and rare-event simulation campaign
G4 Residual cyber risk is 3 or below	Supported at concept stage	TARA with 13 controls	Penetration tests (VER-21, VER-25)

13.2 Regulatory pathway

No regulation today covers a road car that fires thrusters or leaves the ground. The likely path is to keep the two worlds separate.

Domain	Situation	Likely route
Road type approval (US FMVSS, EU, UN)	No standard addresses thrusters, stored compressed air for propulsion or external airbags	Road-legal trims with thrust physically disabled in Road-safe mode; Rocket trim sold as not approved for public road use, as Tesla already states [B9]

Domain	Situation	Likely route
Pressure vessels	Compressed-air propulsion vessels on cars have no dedicated rule	Qualify to the closest standards: UN GTR 13 and UN R134 (hydrogen vessels), UN R110 (CNG) and ISO 11119-3 (composite cylinders)
Aviation (FAA, EASA)	Whether a few-second hop counts as flight is unresolved	Operate only at closed sites under event or test permissions; seek early regulator engagement
Cybersecurity and updates	UN R155 and R156 apply to type-approved vehicles	Apply both to all trims
Track use	Track operators set their own rules	Site procedures, clear zones and marshals, written into the operating domain

13.3 Independent review record

This issue was reviewed against the earlier issues and the public sources. Each finding below has been closed in this document.

ID	Finding	Correction	Where
RV-01	Fill pressure of 600 bar mixed up the 600 L volume with pressure	400 bar adopted, consistent with the 250 kg gas figure	Ch. 5, A-05
RV-02	Burst energy over-stated at 90 MJ	Recalculated: about 60 MJ (about 14 kg TNT)	Ch. 5
RV-03	Launch gas use over-stated at 57–84 kg	Recalculated: 32–46 kg (0.9 s) and 46–67 kg (0.59 s)	Ch. 5
RV-04	Normally closed valves drop the car if power or a computer is lost mid-hop	TSR-21 to TSR-23 added; E/E architecture updated	Ch. 4, 7, 8
RV-05	Both channels trusted the same zone data, a common-mode attack path	Independent zone confirmation CC-13 and AD-09 added	Ch. 4, 10
RV-06	ASIL B for SG6 looked too low for a flight-like manoeuvre	Rationale written; FHA target 10^{-7} per hop added as a quantitative requirement	Ch. 6
RV-07	Coarse FMEDA (Sim 6) not at component level	Superseded by component-level cut-set FMEDA (Sim 7)	Ch. 8
RV-08	First PMHF result was implausibly small because common-cause faults were missing	β-factor common-cause events added	Ch. 8, A-14
RV-09	"SpaceX-built" overstated the sources	Changed to "developed with SpaceX"	Ch. 1, 2
RV-10	Single-source facts presented as confirmed	Each fact tagged with its source; unconfirmed items marked	Ch. 2, App. B
RV-11	Requirements, trace and tables kept in several places could drift	One requirements database; document tables generated from it; automated two-way link check	Ch. 12

13.4 Open issues

ID	Issue	Owner action	Blocks
OI-01	SG6 per-hop risk 3.3×10^{-7} is above the 10^{-7} target	Individually controlled nozzles (TSR-17), better recovery planner (TSR-20)	Release
OI-02	SOTIF residual risk not demonstrated	Validated 6-degree-of-freedom model and scenario campaign (ch. 9)	Release
OI-03	Stuck-closed nozzle during the roll fails about one case in three	Design study for nozzle layout and roll authority	Release
OI-04	Failure rates and β-factors are assumed	Supplier FMEDA data; DFA with test data	Safety case G1
OI-05	Cold-gas effects on valves and seals unknown	Full-discharge cold tests	Safety case G1
OI-06	Regulatory classification of a hop	Early engagement with road and aviation regulators	Market entry

ID	Issue	Owner action	Blocks
OI-07	No requirement stops thrust after touchdown (UCA-8)	Add a landing-detection requirement in the next issue	Next issue
OI-08	A fall from 6 m gives 16.5 g even with external airbags, above the 15 g criterion; the operating domain allows a 6 m apex	Cut the apex limit to 5 m (14 g) or improve the landing system	Operating domain

14. Simulation report

Seven simulations support the analysis. They are deliberately simple, so anyone can read, run and challenge them: about 1,400 lines of Python using NumPy and Matplotlib, all in `sim/`. `python sim/run_all.py` reproduces every number and chart in this chapter in under two minutes on a laptop. Random seeds are fixed, so results are repeatable.

14.1 Common model

Element	Model
Vehicle	Planar rigid body in the roll plane: height, lateral position and roll angle, 1,850 kg, 600 kg·m² (A-01, A-02)
Thrusters	Four lift groups of 9 kN at ±0.35 m and ±0.85 m from the centreline (A-06); first-order valve lag; ±3% thrust scatter
Gas	Adiabatic blowdown of a lumped tank with real-gas density, regulated 40 bar feed, specific impulse falling with gas temperature (A-05, A-07, A-08)
Guidance	Reference hop: climb to 4.5 m in 1.3 s, 360° roll near the apex, descent at up to 4 m/s slowing to 0.8 m/s near the ground
Control	PD attitude and height control with feed-forward; torque-first allocation to the available groups (A-10)
Faults	Stuck-open, stuck-closed or leaking group at any time; detection delay as a parameter
Monitor strategies	None; cut all gas; isolate the faulty group and recover
Outcome	Safe if the car lands with roll below 30° and vertical speed below 5 m/s

14.2 Sim 1: gas budget

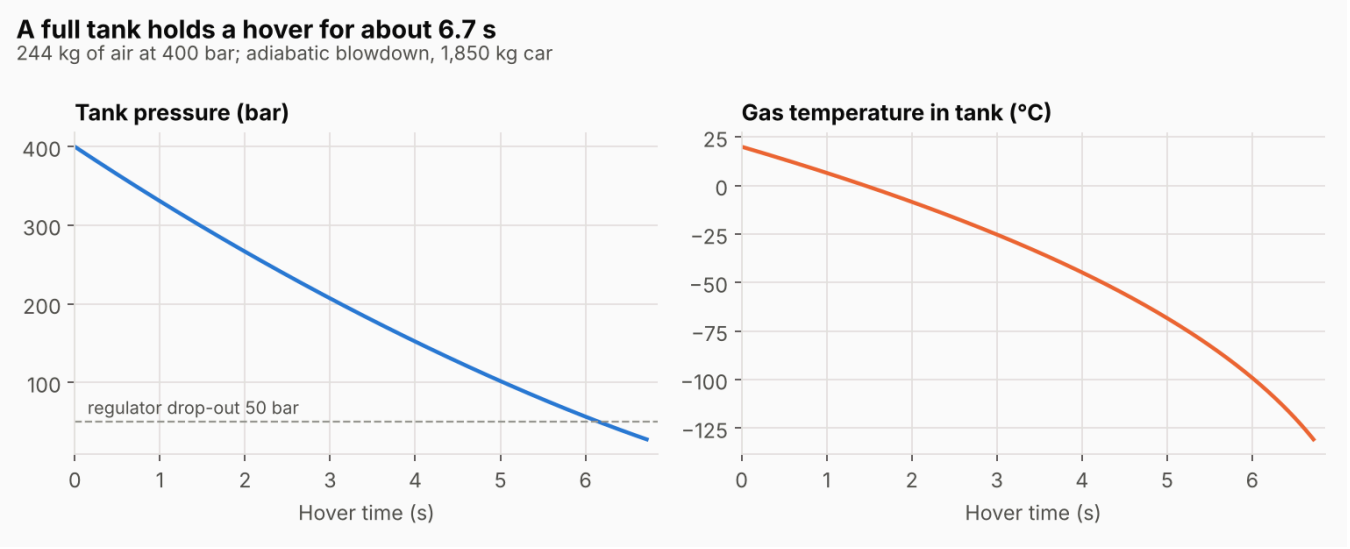


Figure 11. Tank pressure, gas temperature and usable thrust during a full-tank hover.

A full bank of 244 kg at 400 bar holds the car in a hover for 6.7 s. Pressure falls to 28 bar and gas temperature to 142 K. Specific impulse falls from 69 s to 48 s as the gas cools. The reference hop needs 134 kg (55% of the fill). Stored burst energy is 60 MJ, about 14 kg of TNT.

14.3 Sim 2: stuck-open valve, three monitor strategies

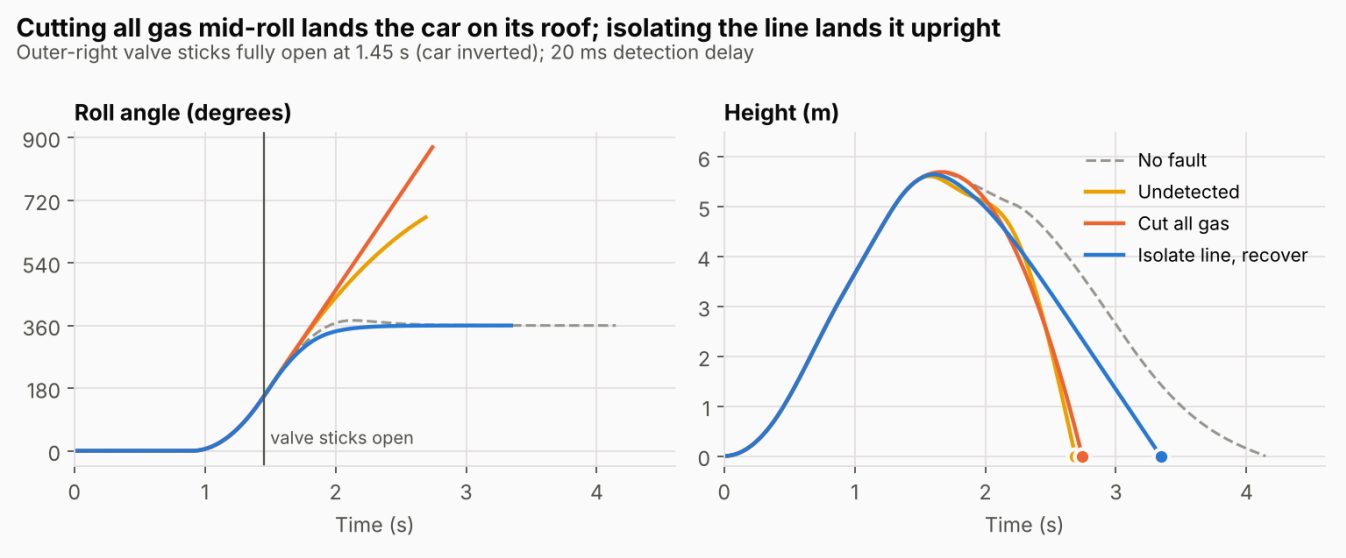


Figure 12. Roll angle and height after an outer lift group sticks open at 1.45 s, when the car is nearly inverted.

Strategy	Touchdown	Roll at touchdown	Vertical speed	Outcome
No fault	4.15 s	0°	1.0 m/s	Safe
Fault not detected	2.69 s	−49°	11.8 m/s	Unsafe
Cut all gas	2.74 s	152° (on the roof)	10.6 m/s	Unsafe
Isolate the line and recover	3.34 s	0°	3.9 m/s	Safe

Cutting all gas, the classic "safe state", lands the car on its roof. This single result drives AD-03 and TSR-12.

14.4 Sim 3: fault tolerant time interval

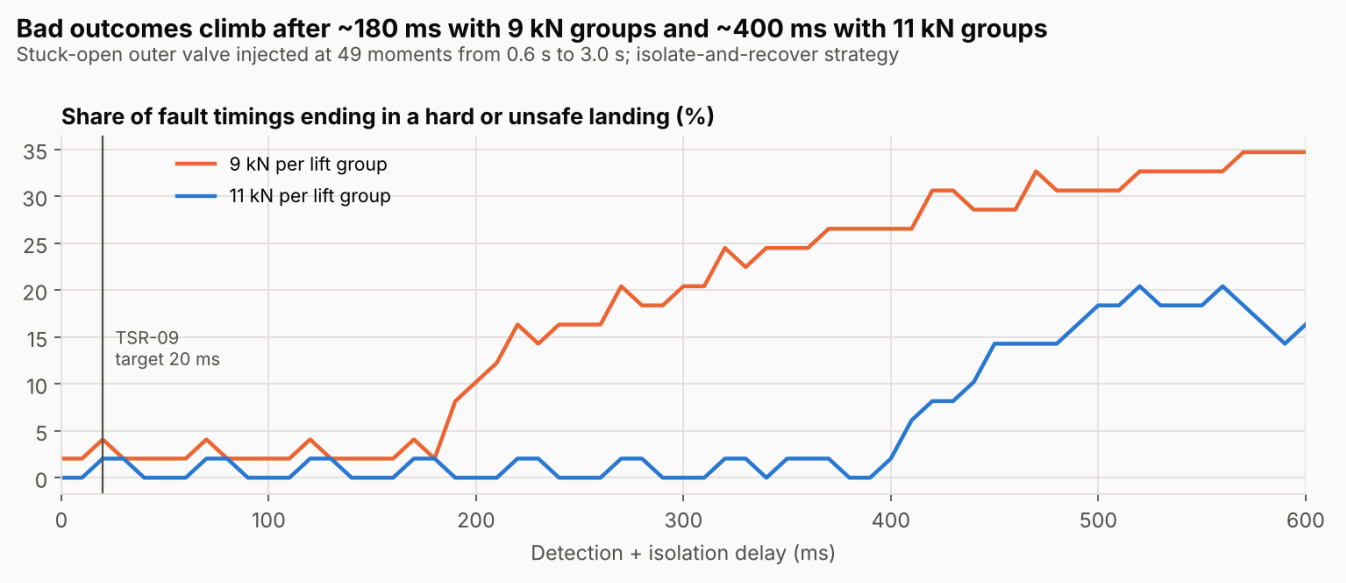


Figure 13. Share of bad outcomes against detection-plus-isolation delay, over fault times from 0.6 s to 3.0 s.

Lift group	Bad at 20 ms	Bad at 100 ms	Bad at 300 ms	Delay where bad outcomes pass 10%
9 kN	4.1%	2.0%	20.4%	200 ms
11 kN	2.0%	0%	0%	440 ms

The FTTI is about 200 ms with 9 kN groups, so a 20 ms budget leaves a tenfold margin. The residual bad outcomes at short delays come from faults in the middle of the roll, where the recovery planner sometimes chooses the longer way back to level. That is the gap recorded against TSR-20. Larger 11 kN groups remove almost all stuck-open failures.

14.5 Sim 4: Monte Carlo over 12,000 hops

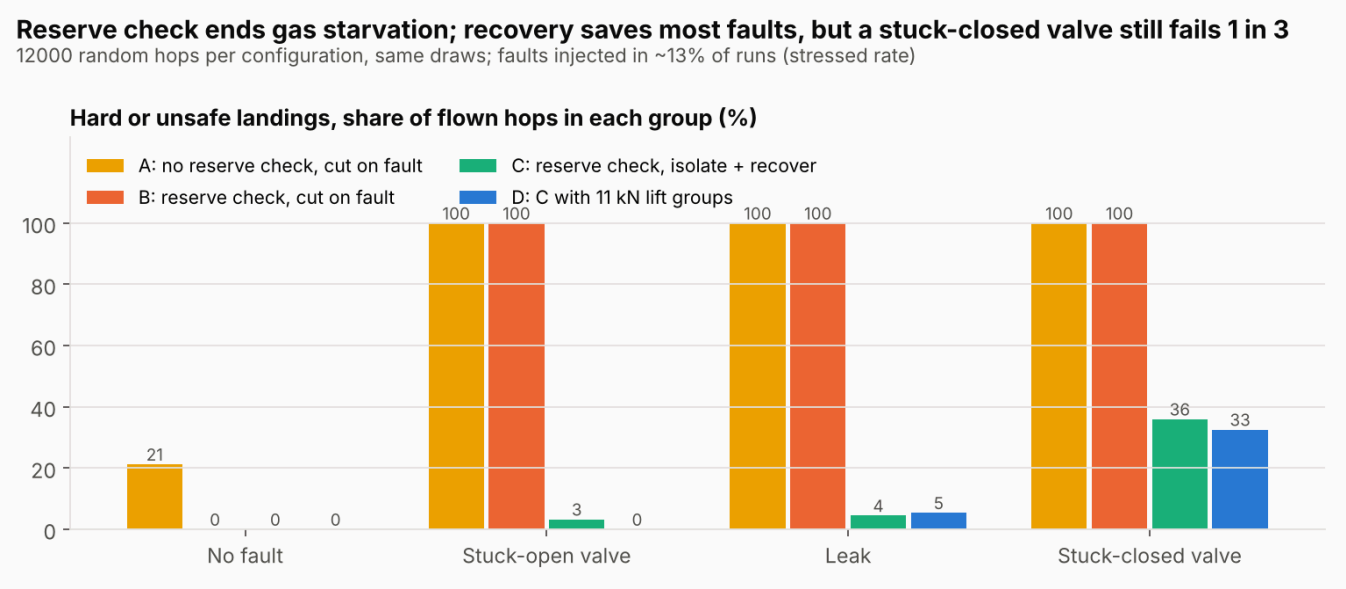


Figure 14. Bad-outcome rates for four architecture options over the same 12,000 random draws.

Each draw varies fill (40–100%), gas temperature, mass, inertia, centre of gravity, gusts and thrust scatter, and injects a stuck-open, leak or stuck-closed fault at an elevated rate so that each fault type gets enough samples.

Option	Hops refused	Bad per flown hop	Bad, no fault	Bad given stuck open	Bad given leak	Bad given stuck closed
A: no reserve check, cut on fault	0%	31.6%	21.2%	100%	100%	100%
B: reserve check, cut on fault	79.1%	12.7%	0%	100%	100%	100%
C: reserve check, isolate and recover	79.1%	1.6%	0%	3.3%	4.5%	36.0%
D: C with 11 kN lift groups	79.1%	1.4%	0%	0%	5.4%	32.6%

The high refusal rate reflects the deliberately wide fill range: the reference hop needs at least 86% fill. In service the reserve check would offer a lower manoeuvre instead of refusing (AD-08). The two key results are that the reserve check removes gas starvation and that isolating and recovering saves almost every stuck-open and leak case. The stuck-closed result of about one in three is the main open design issue (OI-03).

14.6 Sim 5: landing loads

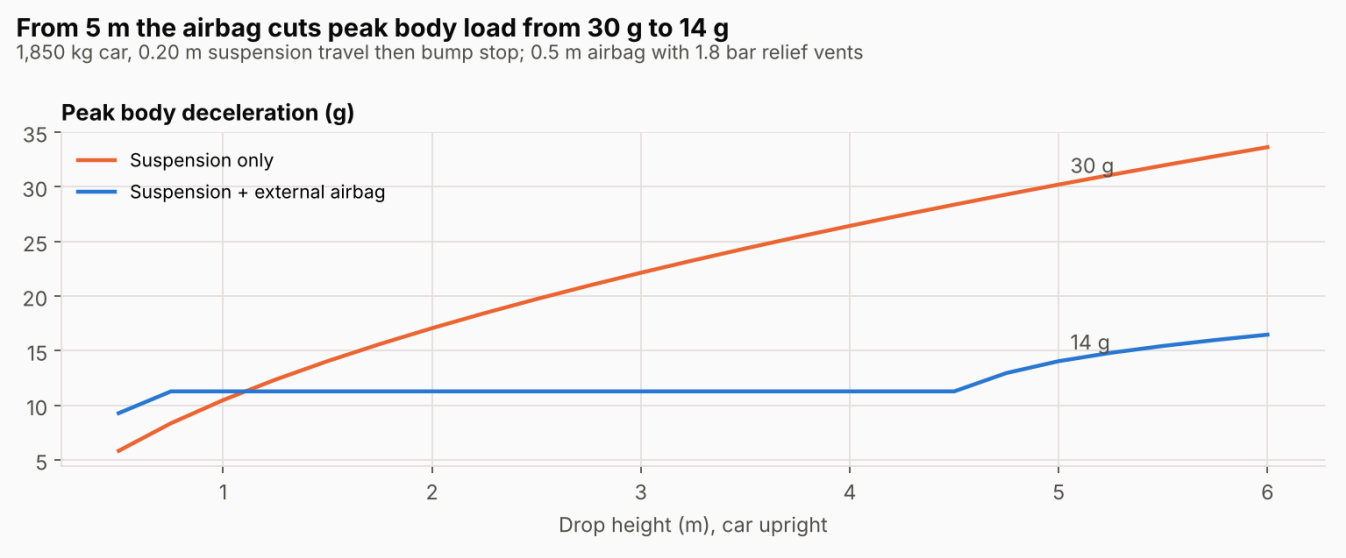


Figure 15. Peak deceleration against drop height, suspension alone and with an external airbag.

Drop height	Suspension only, peak	With airbag, peak
2 m	17.1 g	11.3 g
4 m	26.4 g	11.3 g
5 m	30.2 g	14.0 g
6 m	33.6 g	16.5 g

The airbag's relief vents hold its pressure at 1.8 bar, which caps the peak at about 11 g up to 4 m. Above about 4 m the airbag reaches the end of its stroke and loads rise again. That supports a 5 m limit on apex height (OI-08).

14.7 Sim 6: per-hop risk (first FMEDA superseded)

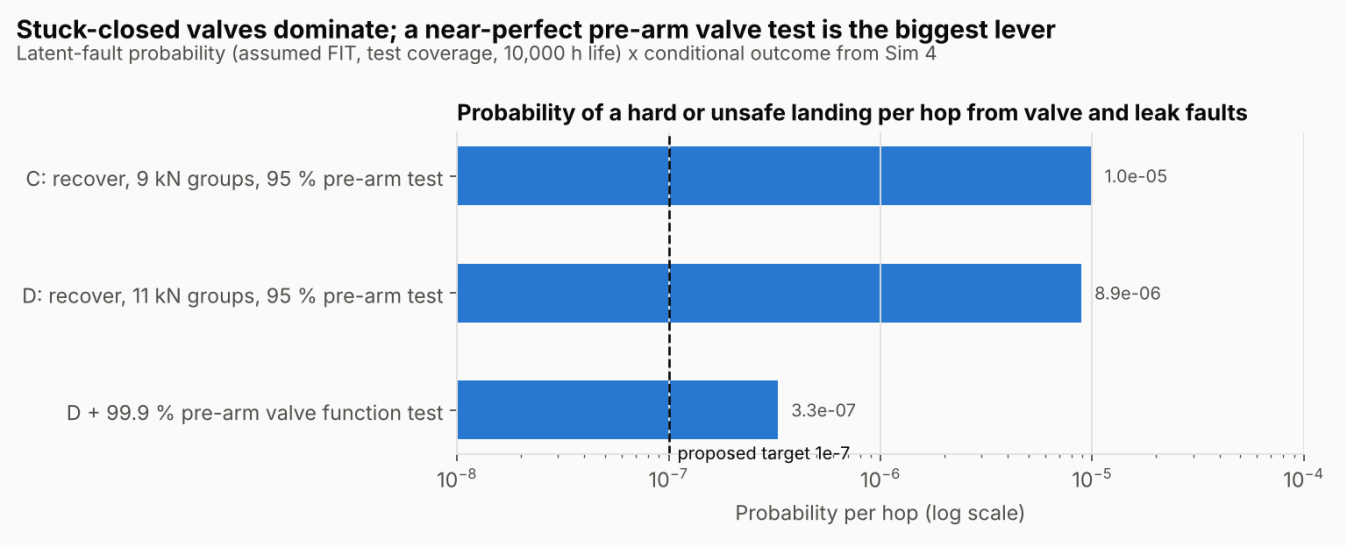


Figure 16. Per-hop probability of a bad landing from random faults, for three design options, against the 10^{-7} target.

Combining Sim 4's conditional outcomes with fault rates and pre-arm test coverage gives the per-hop risk of a bad landing. With 11 kN groups and a 99.9%-coverage pre-arm valve test it is 3.3×10^{-7}, still above the 10^{-7} target; stuck-closed faults dominate. Sim 6 also contains a first, coarse FMEDA, which Sim 7 supersedes.

14.8 Sim 7: component FMEA and cut-set FMEDA

Sim 7 holds the component DFMEA (31 failure modes, chapter 8.2) and solves the SG1 fault tree for its 36 minimal cut sets. It gives SPFM 100%, LFM 98.7% and PMHF 0.006 FIT, with the computers' common-cause term dominating (Figure 6).

14.9 Limits of the models

- Planar model: no pitch or yaw coupling, no lateral wind profile, no plume or ground-effect modelling.
- Lumped tank: no temperature gradients, no heat transfer from vessel walls (this makes the gas colder than reality, a conservative assumption for thrust and a non-conservative one for valve icing).
- Ideal valves with first-order lag; no valve chatter or water hammer.
- Failure rates, coverages and β-factors are assumed (A-11, A-12, A-14).
- No model has been validated against hardware. The results show *relative* merits of architecture options; absolute values need test data.

14.10 Interactive viewer

`viewer/index.html` replays the simulated trajectories in a browser with no installation. It plays 14 precomputed scenarios (the no-fault hop, stuck-open valves with each monitor strategy and several detection delays, stuck-closed valves, leaks, gas starvation and a crosswind gust) and a scan of stuck-closed fault timing that shows the 1.10–1.95 s danger window. A hosted version is the "Roadster Hop Fault Lab".

Appendix A. Assumptions register

Tesla has not published the design. Every number in this study that is not a disclosed fact (chapter 2) rests on one of the assumptions below. Each has a basis and states what real data would replace it. Changing an assumption and running `python sim/run_all.py` and `python tools/build_docs.py` regenerates every affected result.

ID	Parameter	Value	Basis	Replace with
A-01	Vehicle mass	1,850 kg	'4,000+ lb' quoted at the reveal (TechCrunch)	Weighbridge data
A-02	Roll inertia	600 kg m2	Typical low sports car	Inertia rig measurement
A-03	Gas storage volume	600 L	Disclosed by Tesla (L. Moravy)	-
A-04	Gas mass	250 kg reported; 244 kg modelled	InsideEVs, single source	Tesla specification
A-05	Fill pressure	400 bar	Derived: 250 kg in 600 L at 20 C with Z = 1.17	COPV rating
A-06	Thruster count and layout	10 nozzles; four lift groups of 9 kN at +/-0.35 and +/-0.85 m	Count disclosed; layout assumed	Nozzle layout and thrust-stand data
A-07	Thrust coefficient	1.35	Typical cold-gas nozzle with losses	Thrust stand
A-08	Regulation	40 bar feed; full thrust lost below 50 bar tank	Assumed	Regulator characterisation
A-09	Number of COPVs	6 vessels in 3 groups	'Every usable void' implies many vessels	Design data
A-10	Control law	PD with feed-forward, torque-first allocation	Assumed	Production control laws
A-11	Component failure rates	10-300 FIT per part (see component FMEA)	Representative magnitudes	Supplier FMEDA data
A-12	Diagnostic coverage	90-99%; pre-arm test 95% or 99.9%	Assumed	Diagnostic coverage analysis
A-13	Vehicle operating life	10,000 h	Typical automotive mission profile	Programme mission profile
A-14	Common-cause shares	beta = 1% (computers), 2% (master valves)	Typical beta-factor range for diverse redundancy	Dependent failure analysis with test data
A-15	Operating domain limits	Wind < 8 m/s (gusts < 12), -10 to +40 C, dry paved surface	Proposed by this study	Validated limits
A-16	Landing system	Suspension 140 kN/m, 0.20 m travel; airbag 2.6 m2, 0.5 m stroke, 1.8 bar relief	Assumed	Drop tests

Appendix B. Sources

Public sources for the facts in chapter 2, checked between 15 and 18 October 2026. Facts reported by only one outlet are marked.

Ref	Source	Used for
[B1]	TechCrunch: Tesla reveals SpaceX-powered Roadster that flies and falls with style	600 L of compressed air, 10 thrusters, COPVs, remote-operated demonstration, 2028 production
[B2]	The EV Report: Tesla Unveils Production Roadster With SpaceX Cold-Gas Thrusters	Trims, suction system, electric actuators, not approved for public roads
[B3]	InsideEVs: Tesla Just Revealed The Roadster	250 kg of gas, 2G cornering (single source; page unavailable when last checked)
[B4]	Eletric-Vehicles.com: Tesla Unveils Thruster-Equipped Roadster	Acceleration figures, production date
[B5]	Swisher Post: Tesla Roadster unveiled	Acceleration figures, price
[B6]	Benzinga via Webull: Tesla Preps SpaceX Rocket-Powered Flying Roadster Demo	A71 code name (unconfirmed)
[B7]	Tesorb: pre-reveal report on the A71 system	Pressure range (unconfirmed)
[B8]	Electrek: Musk says Roadster's SpaceX package will include rocket thrusters (2018)	2018 SpaceX package, about 10 thrusters
[B9]	Gagadget: Tesla Roadster Gets an October 1 Reveal	Regulatory status

Standards referenced

ISO 26262:2018 (road vehicles, functional safety), ISO 21448:2022 (SOTIF), ISO/SAE 21434:2021 (cybersecurity engineering), UN Regulations No. 155 and 156, UN GTR No. 13, UN Regulations No. 110 and 134, ISO 11119-3, SAE ARP4761A and ARP4754B, AIAG-VDA FMEA Handbook (2019), STPA Handbook (Leveson and Thomas, 2018), GSN Community Standard v3.